

Composition Is Where Obfuscation Robustness Breaks

ANONYMOUS AUTHOR(S)

Fine-tuning a code model on obfuscated programs raises its accuracy on those programs by roughly nineteen points. We ask what that number is made of: competence at reading the program, or competence at the surface a particular transform leaves behind. We separate the two with tests that share no machinery — whether the gain survives being stacked with a transform family the model has never seen, and whether it survives the task being run backwards. The field’s existing answers, inference-time routing, weight-space merging, and training on a breadth of transforms, fail both — merging outright, routing by matching breadth wherever the two are measured together. Breadth composes on stacks built from what it saw and loses that advantage the moment an unseen family enters, a gap significant on every one of six models across four lineages; and every method that fits the forward task buys its gain by losing backward competence on the very same programs. Three further instruments locate what is learned instead: the surface a transform leaves, which is exactly what an obfuscator is free to change. That diagnosis names its own remedy. We anchor the model to the one thing a semantics-preserving transform leaves invariant, its own behaviour on the unobfuscated parent of each program, through a paired-consistency objective that trains on exactly the rows breadth trains on. Re-running the same two tests, the anchored model is the only one whose advantage *grows* as input diverges from training, rising above the clean-code control on stacks containing an unseen family, and the only one whose forward gain costs nothing backwards. Its repair holds on six of eight models and reverses on one, and on the unseen family alone it is as robust as clean-code tuning and no more. Adaptation to obfuscated code buys recombination of what the data covered; anchoring recovers what that costs; neither buys invariance.

1 Introduction

Code obfuscation is widely used to make software difficult to inspect while preserving its intended functionality. In benign settings, developers employ obfuscation to protect intellectual property, implementation details, and proprietary business logic. In adversarial settings, however, malware authors routinely exploit the same techniques to conceal malicious behavior and frustrate program analysis and reverse engineering. Common transformations include identifier renaming, control-flow restructuring, dead-code insertion, expression rewriting, call indirection, etc. Although such transformations are intended to preserve program semantics, they can substantially obscure how those semantics are expressed in the source code, thereby increasing the difficulty of understanding the behavior of malware code [2, 3, 10, 11].

This difficulty is well established for human analysts. Nguyen *et al.* [7], for example, showed that code obfuscation significantly reduces humans’ accuracy in reasoning about program outputs. Such findings are particularly important for malware analysis, where analysts must often infer the behavior of code whose surface structure has been deliberately engineered to resist comprehension.

Large language models (LLMs) offer a promising alternative for automating such reasoning. Models have demonstrated strong capabilities in code generation, summarization, vulnerability detection, and program understanding [1, 4–6, 8]. However, most evaluations are performed on naturally written programs, where meaningful identifiers, familiar control structures, and common coding idioms provide abundant lexical and structural cues. Obfuscated programs remove or distort many of these cues while retaining the underlying computation, creating a useful stress test of whether an LLM actually reasons about program behavior or instead depends on surface regularities.

Recent studies suggest that this distinction matters. Rong *et al.* [?] showed that semantics-preserving obfuscations can cause substantial degradation in LLM program reasoning, including program-output prediction. Similarly, Nikiema *et al.* [9] reported a statistically significant decline in model performance as the complexity of obfuscation increases. Together, these findings indicate that this limitation is especially problematic for security applications: an AI-assisted malware analyzer should not cease to understand a malicious computation merely due to obfuscation.

An important challenge, however, has received considerably less attention: *real obfuscated programs rarely contain only one transformation*. Obfuscators can combine multiple techniques so that, for example, renamed identifiers are embedded inside flattened control flow, rewritten expressions, and additional irrelevant code. Such *stacked obfuscation* creates a fundamentally harder deployment setting. Success on each transformation independently does not necessarily imply success when those transformations occur together. Consequently, simply demonstrating that a model can be adapted to individual obfuscations is insufficient. For practical malware analysis, the learned capability must also *compose*.

This observation naturally suggests several ways to compose knowledge learned from individual transformations. One could train a specialist for each obfuscation and use a *router* to select the appropriate specialist at inference time. Alternatively, one could combine the specialists in parameter space through *model merging*. A third, and seemingly more direct, strategy is to increase *training breadth* by fine-tuning a single model on examples from many individual obfuscations, hoping that knowledge learned separately will recombine when transformations are stacked. We systematically evaluate these three strategies and find that none provides the desired robustness. The learned router is worth about a point over a random routing policy and nothing over breadth training; merging specialized models performs at or below the clean-code-tuned control; and broader training, although effective on stacks assembled from transformations represented during training, introduces a substantial performance penalty on an unseen obfuscation family. Thus, broader exposure can improve recombination within the training distribution while simultaneously making the model *less* robust outside that distribution.

Our analysis reveals why. The apparent robustness obtained from obfuscation-specific adaptation does not primarily arise from learning the semantics that remain invariant under transformation. Instead, models learn regularities associated with the *surface artifacts* left by particular obfuscators. Evidence for this behavior appears across multiple analyses: adaptation transfers poorly between distinct mechanisms within the same obfuscation family; the benefit of broad training is substantially stronger when recognizable identifier cues remain available; and improvements disappear when the same programs are queried in a different reasoning direction. Thus, stacking exposes a deeper problem than merely insufficient training coverage: obfuscation-specific fine-tuning can teach the model how particular transformations *look* without teaching it to consistently recover the behavior that those transformations preserve.

This diagnosis suggests a different principle for robust adaptation. Rather than attempting to compose transformation-specific knowledge, we anchor learning to something that is invariant across all such transformations: the model’s behavior on the corresponding *unobfuscated program*. For every obfuscated training program x_{obf} , we retain its clean parent x_{clean} . A teacher model, tuned for reasoning over clean code, processes x_{clean} , while the student processes x_{obf} . In addition to the ordinary task loss, we minimize the KL divergence between their output distributions:

$$\mathcal{L} = \text{CE}(y \mid x_{\text{obf}}) + \lambda D_{\text{KL}}(p_{\text{teacher}}(\cdot \mid x_{\text{clean}}) \parallel p_{\text{student}}(\cdot \mid x_{\text{obf}})). \quad (1)$$

The first term exposes the student to obfuscated code, whereas the second explicitly encourages it to preserve the behavioral response of a model reasoning over the semantics-equivalent clean program. Importantly, this objective does not require recognizing which obfuscations are present,

selecting among specialists, merging transformation-specific parameters, or recovering the original source code. Instead, it changes the learning target itself: rather than anchoring adaptation to the surface patterns of individual transformations, it anchors the student to the clean-code behavior that those transformations should preserve.

Our empirical results demonstrate the advantage of this *paired-consistency anchoring*. In contrast to routing, model merging, and breadth-only training, the KL-based objective retains the benefits of learning from diverse obfuscations without paying the corresponding penalty on an unseen obfuscation family. At the 7B scale, the anchored model improves over breadth training by 4.59 percentage points on the unseen family while introducing no degradation on clean code. Additional stress tests show that the resulting model continues to perform well under deeper compositions and transfers to a second programming language. At the same time, our analysis carefully bounds the claim: the method does not create a new ability to solve arbitrary unseen obfuscations beyond that of the clean-code teacher. Rather, it prevents obfuscation adaptation from destroying capabilities the model already possesses. This distinction is important: robust reasoning over obfuscated programs requires not only acquiring adaptation gains, but also preserving semantic competence as the surface form of the program changes.

Contributions. This paper makes the following main contributions:

1. A two-test diagnostic for whether obfuscation adaptation generalises or memorises, and its verdict on the existing composition strategies. We ask whether competence learned on individual semantics-preserving transformations survives leaving the distribution it was fit to, in two independent ways: being stacked with a transformation family the model has never seen, and being asked backwards (given a return value, produce a call that yields it). Inference-time routing, weight-space merging, and broad multi-transformation training all fail both. Breadth’s advantage on stacks of seen transformations vanishes once an unseen family enters — a gap significant on six of six models across four lineages — and every method that fits the forward task buys its gain by losing backward competence on the same programs.

2. An empirical characterisation of what adaptation learns instead. With complementary analyses across transformation mechanisms, identifier dependence, and confidence behaviour, we show that obfuscation adaptation predominantly learns the surface regularities a transformation leaves behind rather than an invariance to semantics-preserving change. Because that surface is exactly what an obfuscator is free to alter, the diagnosis names its own remedy.

3. A clean-behaviour anchoring objective, derived from that diagnosis. We introduce a paired-consistency objective that anchors an obfuscated-code student to a clean-code-tuned teacher reading the unobfuscated parent of the same program — the one thing a semantics-preserving transformation leaves invariant. The objective trains on exactly the rows breadth training uses, so breadth is its data-matched control. Across eight models and four lineages it removes breadth’s unseen-family degradation on six and reverses on one, and its clean-code cost is absent on five; we report both qualifiers rather than the unconditional claim.

4. The same two tests, re-run on the anchored model, with the bound stated. On a new divergence ladder of stacks that contain an unseen transformation family — a stimulus absent from prior work — the anchored model’s advantage *grows* as the input diverges from training, rising above the clean-code control where breadth falls, and it is the only supervised-fine-tuning arm whose forward gain costs nothing backwards. It does not acquire backward competence, and on the unseen family alone it is as robust as clean-code tuning and no more: anchoring repairs what adaptation breaks rather than creating new invariance.

2 Research Questions

Fine-tuning a code model on obfuscated programs raises its accuracy on those programs by roughly nineteen points. Every question below asks what that number is made of. Adaptation can raise forward accuracy by teaching the model to *read* the program better, or by teaching it the surface a particular transform leaves behind; only the first deserves to be called robustness. We separate them with two tests that share no machinery, committed before any method was on the table:

- **Composition.** Does competence on single transforms survive being stacked with a transform family the model has never seen?
- **Reversal.** Does it survive the task being run backwards — given a return value, produce a call that yields it? No adapter is trained on this direction, so any backward gain is transferred program understanding rather than task fitting.

RQ1: Do standard adaptation methods generalise? We apply both tests to the field’s existing strategies — inference-time routing over per-transform specialists, weight-space merging of those specialists, and broad multi-transform fine-tuning — treated as baselines rather than proposals, and then locate what they learn instead, because that is what determines the remedy.

RQ2: Does anchoring to clean-code behaviour repair what adaptation breaks? The diagnosis names its own remedy: anchor the model to its own behaviour on the unobfuscated parent of each program. We derive a paired-consistency objective, isolate its mechanism by ablating the teacher, and measure it across eight models and four lineages.

RQ3: Does the anchored model survive the same tests? We re-run RQ1’s two tests on the anchored model and state the bound: a new capability on unseen obfuscation, or a preservation mechanism that protects the teacher’s competence from surface-level change.

3 Setup

3.1 Models

Table 1 lists the panel: eight instruct models from four lineages (Meta, BigCode, Google, IBM), code-specialised and general, 7B to 34B, plus one pretrained checkpoint used only for the one-shot baseline. Every adapter is a LoRA with the same rank and target modules on every model, and one prompt builder serves training, evaluation and attention extraction.

Table 1. **The model panel.** Four lineages, code-specialised and general instruct models, 7B to 34B. *Chat template: system* accepts a system role; *merged* folds the system text into the user turn; *plain* has no template and is used for the one-shot baseline only. Parameter counts are nominal. Every adapter is a LoRA with the same rank and target modules on every model.

Model	Checkpoint	Lineage	Type	Params (B)	Layers	Hidden	Chat template
CodeLlama-7B	codellama/CodeLlama-7b-Instruct-hf	Meta	coder	6.7	32	4096	system
CodeLlama-13B	codellama/CodeLlama-13b-Instruct-hf	Meta	coder	13.0	40	5120	system
CodeLlama-34B	codellama/CodeLlama-34b-Instruct-hf	Meta	coder	33.7	48	8192	system
Llama-3.1-8B	meta-llama/llama-3.1-8B-Instruct	Meta	instruct	8.0	32	4096	system
StarCoder2-15B	bigcode/starcoder2-15b-instruct-v0.1	BigCode	coder	15.5	40	6144	merged
Gemma-3-12B	google/gemma-3-12b-it	Google	instruct	12.2	48	3840	system
CodeGemma-7B	google/codegemma-7b-it	Google	coder	8.5	28	3072	merged
Granite-3.1-8B	ibm-granite/granite-3.1-8b-instruct	IBM	instruct	8.2	40	4096	system
Llama-3.1-8B (base)	meta-llama/llama-3.1-8B	Meta	pretrained	8.0	32	4096	plain

3.2 Dataset

Table 2 summarises the corpus. The split is by program, never by row; three input cases per program are fixed and their gold outputs executed from the clean parent, so the grader is independent of

any transform. Coverage differs by condition because structural transforms decline some programs by design and the unseen family needs enough encodable sites; every paired contrast is computed on the programs common to the cells it compares, fixed before evaluation.

Table 2. **The dataset.** One Python corpus, split by program so no obfuscated variant of a training program can reach evaluation.

Property	Value
Base programs (Python)	2231: apps 1584, cruxeval 543, humaneval 104
Split	by program, seed 17: train 1563 / val 111 / held-out 557
Input cases per program	3, gold output executed from the clean parent
Program length	mean 8.4 LOC, max 57
Held-out eval, clean	557 programs $\times$ 3 cases = 1671 items
Coverage per condition	Tables 3 and 4; structural transforms decline some programs by design, the unseen family needs ≥ 3 encodable sites
Training rows per adapter	one condition's train split $\times$ 3 cases (4689); breadth pools the five seen conditions
Paired contrasts	on the programs common to every cell in the contrast, fixed before evaluation; program-clustered bootstrap, 2,000 resamples
JavaScript	168 held-out programs (504 items), transferred intact; not regenerable on the cluster
Grading	strict normalised exact match, no containment; the backward task executes the produced call against the clean parent

3.3 Obfuscation taxonomy

Single transforms (Table 3) fall into three families: identifier (L1b, L1r, L2), structural (S1, S2, and S2's two halves S3 and S4) and encoding (X1 and its halves). Each is applied to the clean parent and never stacked during training. H1 is the quarantined held-out obfuscator; its evaluation budget was spent under a pre-registered two-pass discipline, and every unseen-family number in this paper is measured on its trainable sibling X1, which reproduces H1's difficulty at $r = 0.9992$ (mean absolute difference 0.48 points). Stacks (Table 4) are a separate namespace built by applying parts to the parent in order; order matters, and no system is trained on a stack. Ten stacks contain only seen transforms (six at depth two, four at depth three to four); six contain the unseen family or one of its halves and are the stimulus RQ4 turns on.

3.4 Systems

Every system is a LoRA on the same base model, evaluated on still-obfuscated code. *Clean LoRA* (tuned_L0) is tuned on unobfuscated programs and is the control throughout; one *specialist* per seen transform; *breadth* (mono_all) is one adapter trained on all five seen transforms pooled; *family* (tuned_X1) is trained on the unseen family's sibling and is the ceiling for that family. *Routing* mixes the specialists with a learned gate (mole_router), its argmax, a fixed uniform gate or a gate frozen at random initialisation; the last two separate what the mixture is worth from what the routing is worth. *Merging* combines the specialists in weight space (TIES, DARE-TIES, DARE-linear) or anchors the merge on the clean adapter. The *anchored* objective (cons_lam3) is Eq. 1 with $\lambda = 3$, a plateau from a sweep on the trainable grid: the teacher is a frozen clean-code-tuned adapter reading the unobfuscated parent, the student never sees the parent, and the objective trains on exactly the rows breadth trains on (26,841 rows, 1,257 steps for both); at $\lambda = 0$ it reduces to breadth identically, which makes breadth the data-matched control. A one-shot in-context baseline completes the set.

Table 3. **Single transforms.** Each is applied to the clean parent and never stacked. *Cap* is the maximum growth factor over the parent above which a program is declined rather than clipped; *programs / items* is held-out coverage. H1 is the quarantined held-out obfuscator whose evaluation budget is spent; X1 is its trainable sibling and carries every unseen-family number.

Code	Family	Transform	Trainable	Cap	Programs / items	Role
L0	none	normalised original: comments and docstrings stripped, 4-space indent	yes	1.2×	557 / 1671	clean reference
L1b	identifier	adversarial renaming of all bindings incl. the entry function, from semantic-inversion tables (max→min) and an unrelated domain vocabulary	yes	2.0×	553 / 1659	ladder
L1r	identifier	random hex renaming of all bindings (v_a3f2, f_9c1e)	yes	2.0×	557 / 1671	ladder
L2	identifier	sequential minification of all bindings (a, b, ..., aa) plus type-annotation stripping	yes	1.5×	557 / 1671	ladder
S1	structural	control-flow flattening of the entry function into a dispatch loop with randomised state ids and shuffled cases	yes	6.0×	416 / 1248	ladder
S2	structural	opaque predicates (computed always-true/false guards) plus never-called dead helpers	yes	4.0×	556 / 1668	ladder
S3	structural	dead-code insertion only: 1–2 never-called module-level helpers (must be ignored)	yes	4.0×	557 / 1671	stacks only
S4	structural	opaque predicates only: 1–3 computed guards at statement boundaries (must be reasoned about)	yes	4.0×	556 / 1668	stacks only
X1	encoding	every string literal through an inline XOR decoder, every + − ^ through a type-guarded algebraic identity, int literals expanded	yes	12.0×	405 / 1215	unseen family
X1m	encoding	X1’s arithmetic half only (mechanisms: [mba])	yes	12.0×	351 / 1053	mechanism ablation
X1s	encoding	X1’s string half only (mechanisms: [str]); site bar 1	yes	12.0×	246 / 738	mechanism ablation
H1	heldout	string encoding (base64 / RC4 string array) plus guarded mixed-boolean-arithmetic rewriting	never	12.0×	–	quarantined

3.5 Measurement

Forward accuracy is strict normalised exact match on the produced output, with no containment matching; the backward task is graded by executing the produced call against the program as the model was shown it — the obfuscated variant, not the clean parent — and accepting any arguments that return the gold value, which is what the prompt asks for. We also report the stricter rate at which the *gold* arguments are recovered. Every contrast is a paired difference on the program subset common to its cells, with a program-clustered bootstrap of 2,000 resamples; bold intervals exclude zero. Decision rules were pre-registered before submission and are reported by their verdicts even when inconclusive. A cell whose format-failure rate exceeds 0.25 measures the prompt contract

Table 4. **Stacked composites.** A separate namespace from the ladder: parts are applied to the clean parent left to right, and order matters because composition does not commute. Size caps were calibrated on a 40-program build at cap 60 ($p95 \times 1.25$) so that the cap is never the binding constraint. No system is ever trained on a stack; the breadth and family adapters see single transforms only.

Code	Parts, in order	Depth	X1 inside	Cap	Prog./items	Group	Purpose
C_L1b_S1	L1b→S1	2	no	8.0×	418 / 1254	seen d2	does L1b's special-ist advantage survive stacking?
C_L1r_S1	L1r→S1	2	no	8.0×	418 / 1254	seen d2	headline identifier ⊗ structural stack
C_S1_L1r	S1→L1r	2	no	8.0×	418 / 1254	seen d2	order control for C_L1r_S1 (renames S1's state variables)
C_L2_S4	L2→S4	2	no	5.0×	556 / 1668	seen d2	identifier ⊗ the reason-about-it half of S2
C_L1r_S3	L1r→S3	2	no	5.0×	557 / 1671	seen d2	identifier ⊗ the ignore-it half of S2
C_S4_S3	S4→S3	2	no	5.0×	556 / 1668	seen d2	positive control: re-constructs S2 by composition
C3_L1r_S3_S4	L1r→S3→S4	3	no	10.0×	556 / 1668	seen d3–4	depth 3, identifier first
C3_S1_S3_S4	S1→S3→S4	3	no	14.0×	395 / 1185	seen d3–4	depth 3, structural only
C3_L1r_S1_S4	L1r→S1→S4	3	no	14.0×	418 / 1254	seen d3–4	depth 3, identifier first, flattening inside
C4_L1r_S1_S3_S4	L1r→S1→S3→S4	4	no	18.0×	418 / 1254	seen d3–4	depth 4
C_L1r_X1	L1r→X1	2	yes	10.0×	405 / 1215	unseen d2	unseen family last, after renaming
C_X1_S1	X1→S1	2	yes	12.0×	321 / 963	unseen d2	unseen material itself flattened
C_S2_X1	S2→X1	2	yes	22.0×	555 / 1665	unseen d2	unseen family applied to dead code too
C_L1r_X1m	L1r→X1m	2	yes	10.0×	351 / 1053	half-unseen d2	half an unseen family (arithmetic)
C_S1_X1s	S1→X1s	2	yes	14.0×	418 / 1254	half-unseen d2	half an unseen family (strings)
C3_L1r_S1_X1	L1r→S1→X1	3	yes	16.0×	417 / 1251	unseen d3	one unseen component at depth 3

rather than the task and enters no mean; untuned format failure measures adherence to that contract, not capability, so screening base models by it is unsound.

4 Every method against every obfuscation

Table 5 is the reference the research questions draw on: the raw accuracy of every method on every condition and stack for CodeLlama-7B, the one model on which every arm exists, forward and, for the ladder, backward. Every number is a strict exact-match accuracy on the held-out programs of that condition; nothing is normalised. The untuned model scores 0.257 on clean code, 0.168–0.207 on the single seen transforms, 0.119 on the unseen family and 0.089–0.186 on the stacks. Every

tuned arm lands between 0.35 and 0.43 on the single seen transforms and their stacks; the columns separate on the unseen family (0.231 breadth, 0.269 clean LoRA, 0.286 anchored, 0.316 family) and on the stacks that contain it. The backward block is read on this model only, because the one-shot inverse template collapses the format on most other panel models; the per-model grids for the whole panel, and the routing and merging arms on every model that has them, are in Appendices B and C.

5 RQ1: Do current adaptation methods generalise?

Adapting a code model to obfuscated input works, and works easily: tuning on output prediction over obfuscated programs gains roughly **19 points** over the untuned model on the transforms it was trained on. We apply the two tests of Section 3 to the field’s existing answers. These are baselines, not proposals. Merging and breadth fail both tests; routing tracks breadth wherever the two are measured together and inherits its verdict.

Routing composes exactly as far as breadth does, and merging does not compose at all (Table 30, Appendix D). On the ten stacks of seen transforms, a learned router beats a *random* gate by +1.34 [+0.48, +2.17]: the mixture is worth the whole gain (+19.79 over the untuned model) and the routing about a point of it. A hard argmax router is indistinguishable from the soft one (+0.03 [−0.13, +0.19]), so a stacked input gives the mixture no blending advantage, and the router matches breadth training, −0.28 [−1.57, +1.09]. The best merge (DARE-TIES) matches the clean-code control, +0.52 [−0.15, +1.20], the others fall 1.7 to 20.6 points below it, while breadth stands +3.67 [+2.02, +5.24] above the same control.

Table 31 (Appendix D) reports the same merge arms measured across the panel.

Breadth is the interesting case. It composes on stacks built from transforms it saw, and the advantage evaporates the moment a stack contains a family it did not. Table 6 states this as a *difference*, because one interval being positive while another is negative is not a test of the difference between them: breadth’s advantage over clean-code tuning is **2.7 to 5.3 points larger** on seen stacks than on stacks containing an unseen family, and the gap is significant on **every one of the six models** we could measure it on, across four lineages. What breadth buys is recombination of what the training data covered, bounded by that coverage.

The reversal test is independent and gives the same answer. Table 32 (Appendix D) reports the *direction ratio* — backward gain divided by forward gain, both against the untuned model — for the same adapters on the same programs asked backwards. **Every method that fits the forward task directly is negative**: the format-only control at −0.30, clean-code tuning at −0.08, breadth at −0.05; these arms bought forward accuracy by losing backward competence on the very same programs. Both tests point at one explanation: what is acquired is competence at *this task on these surfaces*. Three further instruments, sharing no machinery, say which surface. Breadth’s advantage on a stack depends on the stack still containing an *identifier* transform rather than on its depth or the number of mechanisms in it, and the rule defining that split was fixed before the unseen-containing stimulus existed, which makes its second row an out-of-sample test: the difference is +3.39 on all-seen stacks and grows to +5.85 on stacks containing the unseen family. Splitting the held-out family into its two mechanisms, the halves **do not transfer to each other** (+1.49 [−0.95, +3.93] and −0.10 [−2.38, +2.19]) yet **either alone recovers about 90 %** of the whole adapter’s gain on the full family — composable skills would deliver half each. And breadth’s log-probability of the gold answer sits about 4 nats below the clean-code control with the entire deficit in the items it gets wrong: it does not become a worse reader, it becomes a more committed guesser. Adaptation anchors on the surface a transform leaves behind, which is exactly what an obfuscator is free to change. The remedy follows: anchor the model to something the transform *cannot* change — its

Table 5. **Every method against every obfuscation, CodeLlama-7B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.257	100	0.302	117	0.427	166	0.413	161	0.422	164	0.413	161	0.429	167	0.426	166
L1b	0.198	100	0.255	129	0.358	181	0.388	196	0.402	203	0.346	175	0.381	192	0.373	188
L1r	0.207	100	0.265	128	0.378	183	0.387	187	0.395	191	0.360	174	0.383	185	0.381	184
L2	0.202	100	0.272	135	0.382	189	0.378	188	0.398	197	0.354	175	0.386	191	0.375	186
S1	0.168	100	0.216	128	0.379	225	0.389	231	0.400	238	0.370	220	0.395	234	0.376	223
S2	0.193	100	0.215 [†]	-	0.389	202	0.403	208	0.415	215	0.383	198	0.421	218	0.392	203
X1	0.119	100	0.154	129	0.269	225	0.231	194	0.286	239	0.316	265	0.279	234	0.276	231
C_L1b_S1	0.141	100	0.153	108	0.279	197	0.345	244	0.360	255	0.278	197	0.325	230	0.302	214
C_L1r_S1	0.131	100	0.179	137	0.294	224	0.350	267	0.356	272	0.296	226	0.346	264	0.301	230
C_S1_L1r	0.135	100	0.168	125	0.285	211	0.326	242	0.349	259	0.297	220	0.318	236	0.301	223
C_L2_S4	0.176	100	0.239	136	0.355	202	0.385	219	0.392	223	0.343	195	0.383	218	0.363	206
C_L1r_S3	0.172	100	0.201	117	0.360	209	0.380	221	0.387	225	0.350	204	0.391	228	0.368	214
C_S4_S3	0.186	100	0.223 [†]	-	0.392	211	0.403	217	0.422	227	0.382	205	0.415	223	0.398	214
C3_L1r_S3_S4	0.150	100	0.181	121	0.340	227	0.377	251	0.380	254	0.343	228	0.375	250	0.349	233
C3_S1_S3_S4	0.128	100	0.160	124	0.354	276	0.369	288	0.385	300	0.343	267	0.377	293	0.340	264
C3_L1r_S1_S4	0.126	100	0.167	132	0.289	229	0.353	280	0.354	280	0.303	241	0.325	258	0.296	235
C4_L1r_S1_S3_S4	0.089	100	0.122	137	0.278	311	0.336	376	0.356	398	0.292	327	0.334	374	0.290	324
C_L1r_X1	0.102	100	0.121	119	0.184	180	0.199	194	0.222	218	0.254	248	0.189	185	0.193	189
C_X1_S1	0.104	100	0.131	126	0.274	264	0.230	221	0.267	257	0.326	314	0.291	280	0.290	279
C_S2_X1	0.145	100	0.184	126	0.313	215	0.285	196	0.332	228	0.347	239	0.326	224	0.317	218
C_L1r_X1m	0.112	100	0.116	103	0.220	196	0.232	207	0.249	222	0.281	251	0.227	203	0.215	192
C_S1_X1s	0.151	100	0.206	137	0.334	221	0.302	200	0.355	235	0.341	226	0.350	232	0.338	224
C3_L1r_S1_X1	0.104	100	0.127	122	0.219	211	0.222	213	0.250	240	0.271	261	0.242	232	0.232	223
<i>Backward: input prediction, graded by execution</i>																
L0	0.285	100	-	-	0.324	114	0.276	97	0.297	104	0.312	109	0.305	107	0.329	115
L1b	0.265	100	-	-	0.251	95	0.279	105	0.294	111	0.265	100	0.300	113	0.297	112
L1r	0.315	100	-	-	0.327	104	0.276	87	0.293	93	0.327	104	0.312	99	0.352	112
L2	0.282	100	-	-	0.324	115	0.276	98	0.275	97	0.324	115	0.328	116	0.342	121
S1	0.261	100	-	-	0.188 [†]	-	0.262	100	0.280	107	0.280	107	0.255	98	0.308	118
S2	0.330	100	-	-	0.209 [†]	-	0.294	89	0.332	101	0.330	100	0.293	89	0.357	108
X1	0.257	100	-	-	0.221 [†]	-	0.237	92	0.281	110	0.291	113	0.206 [†]	-	0.280	109
C_L1b_S1	0.218	100	-	-	0.163 [†]	-	0.242	111	0.270	124	0.246	113	0.222	102	0.280	129
C_L1r_S1	0.301	100	-	-	0.209 [†]	-	0.255	85	0.270	90	0.300	99	0.227 [†]	-	0.316	105
C_S1_L1r	0.282	100	-	-	0.205	73	0.267	95	0.285	101	0.262	93	0.242	86	0.291	103
C_L2_S4	0.291	100	-	-	0.231 [†]	-	0.260	89	0.283	97	0.315	108	0.309	106	0.349	120
C_L1r_S3	0.296	100	-	-	0.267	90	0.278	94	0.315	106	0.330	112	0.303	102	0.350	118
C_S4_S3	0.315	100	-	-	0.201 [†]	-	0.282	89	0.333	106	0.324	103	0.287	91	0.352	112
C3_L1r_S3_S4	0.303	100	-	-	0.191 [†]	-	0.268	89	0.302	100	0.306	101	0.249 [†]	-	0.336	111
C3_S1_S3_S4	0.292	100	-	-	0.166 [†]	-	0.234	80	0.313	107	0.291	100	0.193 [†]	-	0.321	110
C3_L1r_S1_S4	0.299	100	-	-	0.163 [†]	-	0.229	77	0.278	93	0.262	88	0.181 [†]	-	0.314	105
C4_L1r_S1_S3_S4	0.282	100	-	-	0.183 [†]	-	0.217	77	0.288	102	0.273	97	0.176 [†]	-	0.296	105
C_L1r_X1	0.253	100	-	-	0.210	83	0.200	79	0.250	99	0.245	97	0.176 [†]	-	0.255	101
C_X1_S1	0.251	100	-	-	0.139 [†]	-	0.201	80	0.276	110	0.255	102	0.144 [†]	-	0.285	113
C_S2_X1	0.311	100	-	-	0.153 [†]	-	0.217	70	0.305	98	0.280	90	-	-	0.326	105
C_L1r_X1m	0.250	100	-	-	0.220	88	0.223	89	0.250	100	0.233	93	-	-	0.253	101
C_S1_X1s	0.277	100	-	-	0.144 [†]	-	0.195	71	0.288	104	0.270	98	-	-	0.286	103
C3_L1r_S1_X1	0.263	100	-	-	0.139 [†]	-	0.154 [†]	-	0.231	88	0.254	97	-	-	0.288	109

Table 6. **RQ1: the dissociation, measured within model on one program set.** Breadth (mono_all) against the clean-code control (tuned_L0) on stacks built from *seen* transforms and on stacks containing an *unseen* family. The *difference* is the measured quantity; the two levels are context, and on most models the unseen-side interval straddles zero on its own. Both halves are paired on the 319 programs common to every cell of both groups. Bold intervals exclude zero.

Model	Lineage	Seen stacks	Unseen inside	Difference
CodeLlama-7B	Meta	+3.36 [+1.57, +5.27]	-1.92 [-3.70, -0.14]	+5.28 [+3.20, +7.54]
CodeLlama-34B	Meta	+2.27 [+0.44, +4.09]	-0.77 [-2.62, +1.12]	+3.03 [+0.78, +5.28]
Llama-3.1-8B	Meta	+2.30 [+0.47, +4.15]	-1.05 [-2.72, +0.63]	+3.35 [+1.50, +5.30]
StarCoder2-15B	BigCode	+1.50 [-0.02, +3.00]	-1.19 [-2.93, +0.49]	+2.68 [+0.77, +4.65]
Gemma-3-12B	Google	+2.63 [+0.56, +4.64]	-2.09 [-3.81, -0.42]	+4.72 [+2.62, +6.85]
Granite-3.1-8B	IBM	+3.66 [+1.54, +5.68]	-0.24 [-2.02, +1.57]	+3.91 [+1.41, +6.32]
difference significant on				6 of 6 models

own behaviour on the unobfuscated parent, which exists for every training row by construction. Section 6 is that objective.

Table 7. **RQ2: breadth’s stack gain depends on an identifier transform being present.** Breadth over the clean-code control, split by whether the stack contains an identifier transform, with the *difference* between the groups. The rule defining the split was fixed before the unseen-containing stimulus existed, which makes the second row an out-of-sample test of it. We report the difference because one interval excluding zero while another does not is not a test of the difference between them.

Stimulus	With identifier	Structural only	Difference
All-seen depth-3/4 stacks (in-sample)	+5.00 [+3.02, +6.88]	+1.61 [-1.02, +4.23]	+3.39 [+0.82, +6.16]
Stacks containing the unseen family (out-of-sample)	+1.98 [-0.12, +4.18]	-3.88 [-6.08, -1.71]	+5.85 [+3.10, +8.72]

Takeaway: RQ1

Routing, merging and breadth are baselines, and none survives leaving its training distribution intact. Breadth’s advantage over clean-code tuning is 2.7 to 5.3 points larger on stacks of seen transforms than on stacks containing an unseen family, significant on six of six models across four lineages; and every method that fits the forward task has a negative direction ratio, buying forward accuracy with backward competence on the same programs. Three independent instruments locate what is learned instead: the surface a transform leaves behind — which is exactly what an obfuscator is free to change, and which names the remedy.

6 RQ2: Anchoring to clean-code behaviour

The anchored objective (Section 3, Eq. 1) supervises the student on obfuscated input against a frozen clean-code-tuned teacher reading the unobfuscated parent, on exactly the rows breadth trains on.

The ablation is the contribution. Replacing the clean-code-tuned teacher with an untuned base model makes the arm **collapse** (−8.98 [−11.44, −6.43] against breadth on the unseen family); replacing it with the breadth model makes the arm **inherit breadth’s tax** (−4.20 [−6.27, −2.14] against the clean-code teacher) while matching on seen conditions. Across all three teachers the student lands at roughly the teacher’s unseen-family level plus one point. The seen-condition gain comes from the supervised term and is teacher-independent; the unseen-family behaviour is

distilled from the teacher. The method is “distil from a clean-code-tuned model on the clean parent”, and we name it as such.

Table 8. **RQ3: the three findings across eight models and four lineages**, all read from one evaluation phase. ✓ replicated, ○ inconclusive, × refuted. R1 is breadth’s unseen-family tax (mono_all – tuned_L0 on the held-out family); R2 is anchoring removing it (cons_lam3 – mono_all); R3 is anchoring’s clean-code cost (cons_lam3 – tuned_L0 on clean code). Bold intervals exclude zero.

Model	Lineage	R1 unseen tax	R2 anchoring repair	R3 clean-code cost
CodeLlama-7B	Meta	−3.71 [−5.93, −1.56] ✓	+5.44 [+3.37, +7.41] ✓	−0.54 [−1.98, +1.08] ✓
CodeLlama-13B	Meta	−4.04 [−6.26, −1.89] ✓	+3.87 [+2.06, +5.77] ✓	−0.72 [−2.28, +0.90] ✓
CodeLlama-34B	Meta	−2.72 [−5.02, −0.49] ✓	+3.62 [+1.40, +5.76] ✓	+0.48 [−0.84, +1.80] ✓
Llama-3.1-8B	Meta	−2.72 [−4.70, −0.91] ✓	+3.38 [+1.57, +5.36] ✓	−0.30 [−1.98, +1.26] ✓
StarCoder2-15B	BigCode	−2.72 [−4.94, −0.58] ✓	+2.31 [+0.58, +4.03] ✓	−0.96 [−2.51, +0.60] ✓
Gemma-3-12B	Google	−4.70 [−6.93, −2.55] ✓	+2.64 [+0.91, +4.53] ✓	−1.80 [−3.53, −0.06] ×
CodeGemma-7B	Google	−1.98 [−3.95, +0.16] ○	−0.41 [−2.39, +1.49] ○	−2.04 [−3.77, −0.18] ×
Granite-3.1-8B	IBM	−0.91 [−2.97, +1.15] ○	−1.89 [−3.54, −0.25] ×	−3.41 [−5.39, −1.26] ×
replicated / inconclusive / refuted		6 / 2 / 0	6 / 1 / 1	5 / 0 / 3

What it repairs, and on which models. Table 8 reads all eight models from one evaluation phase; every previously published contrast recomputed through it falls inside the new interval (largest disagreement 0.85 points), so a departure is a property of the model. Breadth’s unseen-family tax has the same sign on every model, six significantly, across four lineages and 7B–34B. The repair holds on six of eight models and **reverses on Granite-3.1-8B**, where the anchored arm is worse than breadth on the unseen family. “Anchoring pays no clean-code cost” holds on five of eight. Every failure falls outside the lineage the findings were discovered on, where all three are four-of-four; with $n = 8$ and vendor confounded with lineage (Fisher’s exact $p = 0.0714$ one-sided) that is a warning that single-lineage evaluation would have hidden all of it, not a claim about lineages.

Takeaway: RQ2

Distilling from a clean-code-tuned teacher on the unobfuscated parent, on exactly the rows breadth trains on, removes breadth’s unseen-family tax on six of eight models and reverses it on one (Granite-3.1-8B), at no clean-code cost on five of eight. The teacher ablation shows where the effect lives: the seen-condition gain is teacher-independent, the unseen-family behaviour is the teacher’s.

7 RQ3: Does the anchored model survive the same tests?

If the diagnosis is right, anchoring to something the transform cannot change should move both tests specifically where the input has left the training distribution.

Composition. Table 9 is the divergence ladder, from all-seen depth-2 stacks (d0) to stacks containing a family the model has never seen (d2, d3); the d2 and d3 stimulus did not exist before this work. Once an unseen component enters, breadth’s advantage over the clean-code control **flips sign**, from +3.49 and +4.15 to −1.71 and −0.35, and anchoring, which merely matches breadth on seen stacks (+0.87 [−0.59, +2.33]), rises **above the control** (+2.13, +3.26) and **above breadth** (+3.84, +3.60). The pre-registered rule required both that anchoring hold and that breadth fall below the control; only the first is established, because the pooled d2 interval spans zero, so by that rule the outcome is UNDECIDED. Split by RQ1’s identifier rule, committed before the first d2 cell was

Table 9. **RQ4: the divergence ladder.** Composites that contain a transform family the model has never seen. Breadth’s advantage flips sign the moment such a component enters, while anchoring rises *above* the clean-code control. The pre-registered rule required both that anchoring hold and that breadth fall below the control; only the first is established, since the pooled interval at d2 straddles zero. CodeLlama-7B, 287 programs common to all six composites, fixed before any system was evaluated.

Level	Stimulus	mono-clean	cons-clean	cons-mono
d0	depth 2, all seen	+3.49 [+2.04, +5.01]	–	+0.87 [–0.59, +2.33]
d1	depth 3–4, all seen	+4.15 [+2.37, +5.91]	+5.02 [+3.41, +6.69]	+0.87 [–0.59, +2.33]
d2	depth 2, unseen inside	–1.71 [–3.60, +0.31]	+2.13 [+0.66, +3.60]	+3.84 [+1.90, +5.70]
d3	depth 3, unseen inside	–0.35 [–2.67, +1.86]	+3.26 [+0.81, +5.92]	+3.60 [+1.40, +5.81]

written, identifier-containing stacks sit at +1.98 [–0.12, +4.18] and structural-containing stacks at –3.88 [–6.08, –1.71]: breadth is broken where the stack leaves it no identifier surface to key on. Across the panel (Table 33, Appendix D) the anchoring advantage on unseen-containing stacks replicates on five of eight models, is inconclusive on three and refuted on none; family exposure is significant on all eight and is the largest effect in the table.

Reversal. In Table 32, anchoring’s direction ratio is +0.03: the only arm in the supervised-fine-tuning family whose forward gain cost nothing backwards. It does **not** acquire backward competence (+0.62 [–1.56, +2.81] against the untuned model); the claim is that every existing method pays backward competence for its forward gain and anchoring is the first that does not. The one arm that gains in both directions is trained on the unseen family’s sibling (+1.63 [+0.11, +3.20] backwards): exposure to a family transfers in a way exposure to transforms does not.

The bound. On the unseen family alone the anchored model is as robust as clean-code tuning and no more (–0.30, –0.33, +0.74 against the control at 7B, 13B, 34B; seventh of thirty-three arms), and every arm that beats it there was trained on the family’s sibling. Anchoring buys breadth’s recombination benefit at little cost, extends it to composed inputs the model has never seen, and stops paying the backward price – and it does not add the capability breadth lacks. It is a repair of what adaptation breaks, not a route to invariance.

Takeaway: RQ3

The anchored model’s advantage appears only where the input diverges from training: level with breadth on seen stacks, above the clean-code control and above breadth once an unseen family enters (five of eight models replicate, none refute; the strict pre-registered rule is undecided), and a direction ratio of +0.03 where every other supervised arm is negative. On the unseen family alone it is as robust as clean-code tuning and no more: a repair of what adaptation breaks, not a route to invariance.

8 Related Work

This paper joins three distinct conversations in the literature. First, we differentiate from the deobfuscation (DOBF) lineage by strictly evaluating code that remains obfuscated, measuring semantic competence under transformation rather than recovery capability. Second, in the domain of model merging and modular adaptation, we contribute a negative result supported by a highly rigorous random-gate control, warning that task-vector geometry is often dominated by initialisation rather than genuine knowledge. Finally, we contribute to the growing literature on robustness to semantics-preserving transformations and direction specificity in code understanding.

9 Threats to Validity

The held-out family is a proxy. Every unseen-family number is measured on X1, a trainable sibling of the quarantined family H1, whose budget was spent under a pre-registered two-pass discipline; X1 reproduces H1’s difficulty at $r = 0.9992$ and transfer between them was lossless (0.08 points) for arms trained on the sibling. It is a single family pair; a second hard family with a different surface would settle whether “family” means “surface family”. **Coverage is not uniform.** Conditions cover different program sets; every contrast runs on a subset recorded before evaluation, and where the intersection moves an estimate we report both. **Eight models is a small sample for a claim about models,** and Section 6 declines to make one; the panel’s strongest non-lineage model is also the one an untuned capability screen would have discarded. **Coverage of the baselines.** Merging is measured forward only, on four models (Table 31); routing is measured on seen stacks on one model, so its verdict rests on matching breadth where the two are measured together; the direction ratio is one model deep, because the one-shot inverse template collapses the format on most other panel models. **Instrument failures are reported, not omitted.** The causal attention instrument is inert (an identifier knockout moves the gold log-probability by under 0.5 % everywhere), so the attention account is correlational; the weakest-link test behind the shared-surface reading in Section 5 returned undecided. **Single language.** Every headline is Python; the JavaScript corpus exists but cannot be regenerated on our cluster.

10 Data Availability

Data/code is available in project’s website [].

References

- [1] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. 2021. Evaluating Large Language Models Trained on Code. *arXiv:2107.03374* [cs.LG] <https://arxiv.org/abs/2107.03374>
- [2] Christian Collberg, Clark Thomborson, and Douglas Low. 1997. A Taxonomy of Obfuscating Transformations. *Technical Report* (1997).
- [3] Christian Collberg, Clark Thomborson, and Douglas Low. 1998. Manufacturing cheap, resilient, and stealthy opaque constructs. In *Proceedings of the 25th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages* (San Diego, California, USA) (POPL '98). Association for Computing Machinery, New York, NY, USA, 184–196. doi:10.1145/268946.268962
- [4] Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, Ming Gong, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, and Ming Zhou. 2020. CodeBERT: A Pre-Trained Model for Programming and Natural Languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 1536–1547. doi:10.18653/v1/2020.findings-emnlp.139
- [5] Daya Guo, Shuo Ren, Shuai Lu, Zhangyin Feng, Duyu Tang, Shujie LIU, Long Zhou, Nan Duan, Alexey Svyatkovskiy, Shengyu Fu, Michele Tufano, Shao Kun Deng, Colin Clement, Dawn Drain, Neel Sundaresan, Jian Yin, Daxin Jiang, and Ming Zhou. 2021. GraphCode(BERT): Pre-training Code Representations with Data Flow. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=jLoC4ez43PZ>
- [6] Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, Rémi Leblond, Tom Eccles, James Keeling, Felix Gimeno, Agustin Dal Lago, Thomas Hubert, Peter Choy, Cyprien de Masson d'Autume, Igor Babuschkin, Xinyun Chen, Po-Sen Huang, Johannes Welbl, Sven Gowal, Alexey Cherepanov, James Molloy, Daniel J. Mankowitz, Esme Sutherland Robson, Pushmeet Kohli, Nando de Freitas, Koray Kavukcuoglu, and Oriol Vinyals. 2022. Competition-level code generation with AlphaCode. *Science* 378, 6624 (Dec. 2022), 1092–1097. doi:10.1126/science.abq1158
- [7] Anh H. N. Nguyen, Jack Le, Ilse Lahnstein Coronado, and Tien N. Nguyen. 2026. The Effect of Code Obfuscation on Human Program Comprehension. *arXiv:2603.07668* [cs.SE] <https://arxiv.org/abs/2603.07668>
- [8] Erik Nijkamp, Bo Pang, Hiroaki Hayashi, Lifu Tu, Huan Wang, Yingbo Zhou, Silvio Savarese, and Caiming Xiong. 2023. CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis. *arXiv:2203.13474* [cs.LG] <https://arxiv.org/abs/2203.13474>
- [9] Serge Lionel Nikiema, Jordan Samhi, Abdoul Kader Kaboré, Jacques Klein, and Tegawendé F. Bissyandé. 2025. The Code Barrier: What LLMs Actually Understand? *arXiv:2504.10557* [cs.SE] <https://arxiv.org/abs/2504.10557>
- [10] Philip O'Kane, Sakir Sezer, and Kieran McLaughlin. 2011. Obfuscation: The Hidden Malware. *IEEE Security and Privacy* 9, 5 (2011), 41–47. doi:10.1109/MSP.2011.98
- [11] Daniel Votipka, Seth Rabin, Kristopher Micinski, Jeffrey S. Foster, and Michelle L. Mazurek. 2019. An Observational Investigation of Reverse Engineers' Process and Mental Models. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3290607.3313040

A Per-model absolute tables

Table 5 in the body is CodeLlama-7B, the model on which every arm exists. The same table for the other seven models follows; a cell that has not been evaluated renders as ‘–’, and the mixture column exists only where a router gate has been trained.

B Per-model master grids

One table per panel model: every condition and every stack, forward and backward, with the raw accuracy, the change against the untuned model, and the percentage of the untuned model’s clean-code accuracy. Cells marked *fmt* exceed the pre-registered format-failure gate (0.25) and enter no mean; the gate is applied per cell, which is stricter than the pooled registration. The untuned CodeLlama-34B sits at 0.254 on clean code, a hair over the bar, which gates its percentage column while its deltas stand. Rows marked ‘–’ are cells not yet run at the time of generation.

C Routing and merging grids

Every routing and merging system on every condition and stack that has it, for the models on which those arms exist (routing: CodeLlama-7B; merging: CodeLlama-7B, Granite-3.1-8B, Llama-3.1-8B, StarCoder2-15B), with the same three numbers as Appendix B. The routing arms are MoLE mixtures of the eight per-transform experts; *uniform* and *random* gates separate the mixture’s worth from the routing’s. Every routing cell ran through the mixture engine.

D Supplementary tables

The contrast tables referenced from the research questions that are not the one table each question carries in the body: routing and merging on stacks (Table 30), merging across the panel (Table 31), the direction ratio (Table 32), the anchoring advantage by model (Table 33), and the cross-model summary in relative terms (Table 34).

Table 10. **Every method against every obfuscation, CodeLlama-13B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.252	100	0.278 [†]	-	0.469	186	0.446	177	0.462	183	0.463	184	0.477	189	0.477	189
L1b	0.222	100	0.220 [†]	-	0.397	179	0.423	190	0.438	198	0.379	171	0.426	192	0.400	180
L1r	0.225	100	0.233 [†]	-	0.415	184	0.420	186	0.439	195	0.407	181	0.428	190	0.425	189
L2	0.228	100	0.244 [†]	-	0.407	178	0.417	183	0.432	189	0.401	176	0.424	186	0.421	185
S1	0.213	100	0.216	101	0.408	191	0.413	194	0.435	204	0.410	192	0.435	204	0.435	204
S2	0.184	100	0.187 [†]	-	0.422	229	0.445	241	0.469	254	0.422	229	0.453	246	0.441	239
X1	0.149	100	0.141 [†]	-	0.294	197	0.254	170	0.292	196	0.353	237	-	-	0.306	205
C_L1b_S1	0.183	100	0.159	87	0.318	174	0.367	201	0.375	205	0.320	175	-	-	0.329	180
C_L1r_S1	0.175	100	0.175	100	0.318	182	0.371	212	0.376	215	0.342	196	-	-	0.356	204
C_S1_L1r	0.172	100	0.172	100	0.315	184	0.359	209	0.363	212	0.350	204	-	-	0.341	199
C_L2_S4	0.200	100	0.187 [†]	-	0.389	195	0.422	211	0.432	216	0.385	192	-	-	0.404	202
C_L1r_S3	0.186	100	0.194 [†]	-	0.407	218	0.409	220	0.434	233	0.399	214	-	-	0.417	224
C_S4_S3	0.192	100	0.211 [†]	-	0.424	221	0.436	227	0.459	239	0.433	226	-	-	0.443	231
C3_L1r_S3_S4	0.166	100	0.169 [†]	-	0.382	230	0.410	247	0.429	258	0.382	230	-	-	0.394	237
C3_S1_S3_S4	0.147	100	0.158	107	0.367	250	0.408	278	0.423	288	0.384	261	-	-	0.387	263
C3_L1r_S1_S4	0.163	100	0.156	96	0.314	193	0.367	225	0.367	225	0.349	214	-	-	0.346	212
C4_L1r_S1_S3_S4	0.117	100	0.117	100	0.300	258	0.353	303	0.360	309	0.331	284	-	-	0.320	275
C_L1r_X1	0.123	100	0.115 [†]	-	0.199	162	0.213	173	0.236	192	0.284	232	-	-	0.217	177
C_X1_S1	0.138	100	0.135	98	0.295	214	0.270	195	0.307	222	0.341	247	-	-	0.310	224
C_S2_X1	0.160	100	0.167 [†]	-	0.335	209	0.334	208	0.356	222	0.376	234	-	-	0.353	220
C_L1r_X1m	0.138	100	0.127 [†]	-	0.246	179	0.248	180	0.279	202	0.319	232	-	-	0.251	182
C_S1_X1s	0.179	100	0.166	93	0.364	204	0.345	193	0.360	201	0.368	206	-	-	0.376	210
C3_L1r_S1_X1	0.128	100	0.107 [†]	-	0.233	182	0.234	183	0.249	194	0.291	228	-	-	0.251	196
<i>Backward: input prediction, graded by execution</i>																
L0	0.285	100	-	-	0.269	94	0.066 [†]	-	0.231	81	0.086 [†]	-	0.232 [†]	-	0.339	119
L1b	0.277	100	-	-	0.285	103	0.066 [†]	-	0.253	91	0.133 [†]	-	-	-	0.318	115
L1r	0.327	100	-	-	0.347	106	0.051 [†]	-	0.229	70	0.068 [†]	-	-	-	0.366	112
L2	0.279	100	-	-	0.232 [†]	-	0.026 [†]	-	0.206	74	0.091 [†]	-	-	-	0.318	114
S1	0.304	100	-	-	0.341	112	0.059 [†]	-	0.199	65	0.156 [†]	-	-	-	0.363	119
S2	0.293	100	-	-	0.317	108	0.038 [†]	-	0.238	81	0.139 [†]	-	-	-	0.365	125
X1	0.258	100	-	-	0.274	106	0.016 [†]	-	0.219	85	0.083 [†]	-	-	-	0.298	115
C_L1b_S1	0.253	100	-	-	0.326	129	0.059 [†]	-	0.201	79	0.195 [†]	-	-	-	0.341	135
C_L1r_S1	0.325	100	-	-	0.360	111	0.070 [†]	-	0.203	63	0.124 [†]	-	-	-	0.368	114
C_S1_L1r	0.323	100	-	-	0.351	109	0.086 [†]	-	0.230	71	0.118 [†]	-	-	-	0.376	117
C_L2_S4	0.274	100	-	-	0.186 [†]	-	0.020 [†]	-	0.168 [†]	-	0.061 [†]	-	-	-	0.321	117
C_L1r_S3	0.305	100	-	-	0.361	118	0.041 [†]	-	0.220	72	0.115 [†]	-	-	-	0.366	120
C_S4_S3	0.293	100	-	-	0.326	111	0.026 [†]	-	0.239	82	0.125 [†]	-	-	-	0.363	124
C3_L1r_S3_S4	0.310	100	-	-	0.337	109	0.032 [†]	-	0.224	72	0.080 [†]	-	-	-	0.353	114
C3_S1_S3_S4	0.295	100	-	-	0.329	112	0.035 [†]	-	0.235	80	0.150 [†]	-	-	-	0.347	118
C3_L1r_S1_S4	0.316	100	-	-	0.353	112	0.061 [†]	-	0.211	67	0.137 [†]	-	-	-	0.370	117
C4_L1r_S1_S3_S4	0.310	100	-	-	0.333	107	0.049 [†]	-	0.222	71	0.134 [†]	-	-	-	0.353	114
C_L1r_X1	0.281	100	-	-	0.254	91	0.016 [†]	-	0.170 [†]	-	0.065 [†]	-	-	-	0.266	95
C_X1_S1	0.289	100	-	-	0.264	91	0.007 [†]	-	0.201 [†]	-	0.104 [†]	-	-	-	0.319	110
C_S2_X1	0.271	100	-	-	0.279	103	0.008 [†]	-	0.204 [†]	-	0.092 [†]	-	-	-	0.329	122
C_L1r_X1m	0.291	100	-	-	0.259	89	0.020 [†]	-	0.179 [†]	-	0.072 [†]	-	-	-	0.283	97
C_S1_X1s	0.301	100	-	-	0.290	96	0.025 [†]	-	0.199 [†]	-	0.093 [†]	-	-	-	0.317	105
C3_L1r_S1_X1	0.304	100	-	-	0.293	97	0.026 [†]	-	0.185 [†]	-	0.071 [†]	-	-	-	0.313	103

Table 11. **Every method against every obfuscation, CodeLlama-34B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.254 [†]	-	0.178 [†]	-	0.520	-	0.494	-	0.525	-	0.502	-	-	-	0.527	-
L1b	0.205	100	0.077 [†]	-	0.417	204	0.469	229	0.494	241	0.401	196	-	-	0.429	209
L1r	0.220	100	0.117 [†]	-	0.469	214	0.457	208	0.487	222	0.459	209	-	-	0.467	213
L2	0.231	100	0.109 [†]	-	0.466	202	0.465	202	0.478	207	0.456	198	-	-	0.463	201
S1	0.222	100	0.079 [†]	-	0.472	212	0.467	210	0.496	223	0.461	208	-	-	0.456	205
S2	0.191	100	0.040 [†]	-	0.484	253	0.484	254	0.525	275	0.478	251	-	-	0.484	253
X1	0.135	100	0.050 [†]	-	0.325	241	0.298	221	0.334	248	0.381	282	-	-	0.328	243
C_L1b_S1	0.156	100	0.033 [†]	-	0.322	206	0.399	255	0.421	269	0.336	215	-	-	0.331	212
C_L1r_S1	0.184	100	0.063 [†]	-	0.351	191	0.393	214	0.427	233	0.366	199	-	-	0.362	197
C_S1_L1r	0.180	100	0.053 [†]	-	0.367	204	0.405	225	0.425	237	0.379	211	-	-	0.377	210
C_L2_S4	0.191	100	0.045 [†]	-	0.431	226	0.454	238	0.478	251	0.431	226	-	-	0.436	229
C_L1r_S3	0.169	100	0.041 [†]	-	0.451	266	0.454	268	0.482	284	0.437	258	-	-	0.453	267
C_S4_S3	0.199	100	0.041 [†]	-	0.482	242	0.487	244	0.521	261	0.476	239	-	-	0.488	245
C3_L1r_S3_S4	0.148	100	0.017 [†]	-	0.416	281	0.452	305	0.473	319	0.424	286	-	-	0.425	287
C3_S1_S3_S4	0.128	100	0.042 [†]	-	0.428	336	0.454	356	0.472	370	0.440	345	-	-	0.442	346
C3_L1r_S1_S4	0.147	100	0.041 [†]	-	0.338	230	0.390	266	0.412	280	0.359	245	-	-	0.354	241
C4_L1r_S1_S3_S4	0.105	100	0.033 [†]	-	0.316	302	0.372	356	0.413	395	0.354	338	-	-	0.351	336
C_L1r_X1	0.115	100	0.044 [†]	-	0.226	196	0.249	216	0.278	241	0.275	239	-	-	0.236	205
C_X1_S1	0.125	100	0.064 [†]	-	0.317	254	0.290	233	0.339	272	0.372	298	-	-	0.317	254
C_S2_X1	0.176	100	0.099 [†]	-	0.372	211	0.355	202	0.403	229	0.395	225	-	-	0.382	217
C_L1r_X1m	0.128	100	0.052 [†]	-	0.261	204	0.275	214	0.294	229	0.317	247	-	-	0.279	217
C_S1_X1s	0.176	100	0.077 [†]	-	0.390	221	0.382	217	0.415	235	0.411	233	-	-	0.387	219
C3_L1r_S1_X1	0.118	100	0.048 [†]	-	0.256	218	0.262	222	0.297	252	0.293	249	-	-	0.267	227
<i>Backward: input prediction, graded by execution</i>																
L0	0.363	100	-	-	0.260	72	0.200 [†]	-	0.284	78	0.269	74	-	-	0.382	105
L1b	0.327	100	-	-	0.253	77	0.174 [†]	-	0.287	88	0.227	69	-	-	0.360	110
L1r	0.339	100	-	-	0.255	75	0.135 [†]	-	0.256	76	0.241	71	-	-	0.373	110
L2	0.357	100	-	-	0.280	78	0.179 [†]	-	0.276	77	0.268	75	-	-	0.378	106
S1	0.323	100	-	-	0.305	95	0.178 [†]	-	0.264	82	0.246	76	-	-	0.390	121
S2	0.350	100	-	-	0.354	101	0.219 [†]	-	0.306	87	0.278	79	-	-	0.437	125
X1	0.303	100	-	-	0.253	83	0.118 [†]	-	0.249	82	0.207	68	-	-	0.338	112
C_L1b_S1	0.257	100	-	-	0.272	106	0.144 [†]	-	0.258	100	0.178 [†]	-	-	-	0.358	139
C_L1r_S1	0.304	100	-	-	0.285	94	0.132 [†]	-	0.200	66	0.254	84	-	-	0.375	123
C_S1_L1r	0.292	100	-	-	0.307	105	0.142 [†]	-	0.216	74	0.264	90	-	-	0.371	127
C_L2_S4	0.344	100	-	-	0.291	85	0.170 [†]	-	0.273	79	0.258	75	-	-	0.384	111
C_L1r_S3	0.333	100	-	-	0.327	98	0.167 [†]	-	0.256	77	0.269	81	-	-	0.419	126
C_S4_S3	0.352	100	-	-	0.345	98	0.218 [†]	-	0.301	86	0.276	79	-	-	0.442	126
C3_L1r_S3_S4	0.325	100	-	-	0.330	102	0.182 [†]	-	0.251	77	0.272	84	-	-	0.427	132
C3_S1_S3_S4	0.330	100	-	-	0.323	98	0.208 [†]	-	0.288	87	0.270	82	-	-	0.413	125
C3_L1r_S1_S4	0.291	100	-	-	0.311	107	0.121 [†]	-	0.199	68	0.271	93	-	-	0.381	131
C4_L1r_S1_S3_S4	0.283	100	-	-	0.333	118	0.164 [†]	-	0.214	75	0.270	95	-	-	0.386	136
C_L1r_X1	0.275	100	-	-	0.240	87	0.060 [†]	-	0.162 [†]	-	0.195	71	-	-	0.299	109
C_X1_S1	0.298	100	-	-	0.275	92	0.089 [†]	-	0.216 [†]	-	0.228	77	-	-	0.345	116
C_S2_X1	0.378	100	-	-	0.366	97	0.196 [†]	-	0.288	76	0.233	62	-	-	0.414	110
C_L1r_X1m	0.269	100	-	-	0.230	86	0.077 [†]	-	0.158 [†]	-	0.189	70	-	-	0.308	114
C_S1_X1s	0.346	100	-	-	0.302	87	0.141 [†]	-	0.268 [†]	-	0.250	72	-	-	0.418	121
C3_L1r_S1_X1	0.304	100	-	-	0.296	97	0.084 [†]	-	0.188 [†]	-	0.237	78	-	-	0.357	118

Table 12. **Every method against every obfuscation, Llama-3.1-8B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.258	100	0.320	124	0.448	174	0.426	165	0.445	172	0.459	178	0.475	184	0.462	179
L1b	0.222	100	0.276	124	0.361	163	0.387	174	0.407	183	0.369	166	0.422	190	0.386	174
L1r	0.242	100	0.284	118	0.387	160	0.387	160	0.405	167	0.378	156	0.407	168	0.398	164
L2	0.215	100	0.278	130	0.391	182	0.386	180	0.409	190	0.387	180	0.425	198	0.396	184
S1	0.213	100	0.256	120	0.395	185	0.391	183	0.413	194	0.409	192	0.428	201	0.412	193
S2	0.205	100	0.272	133	0.413	201	0.413	201	0.445	217	0.438	213	0.456	222	0.437	213
X1	0.132	100	0.161	122	0.249	189	0.222	168	0.255	194	0.332	252	0.280	212	0.266	202
C_L1b_S1	0.152	100	0.185	122	0.279	184	0.338	223	0.368	243	0.305	201	0.369	243	0.307	203
C_L1r_S1	0.162	100	0.220	136	0.303	187	0.352	217	0.363	224	0.314	194	0.366	226	0.328	202
C_S1_L1r	0.159	100	0.208	131	0.277	174	0.338	213	0.354	223	0.305	192	0.347	219	0.306	193
C_L2_S4	0.205	100	0.257	125	0.349	170	0.386	188	0.401	196	0.350	171	0.413	201	0.383	187
C_L1r_S3	0.198	100	0.266	134	0.367	185	0.382	193	0.398	201	0.369	186	0.407	205	0.387	195
C_S4_S3	0.214	100	0.269	126	0.421	197	0.411	192	0.444	208	0.431	202	0.459	215	0.437	204
C3_L1r_S3_S4	0.183	100	0.233	128	0.329	180	0.374	205	0.391	214	0.363	198	0.399	218	0.370	202
C3_S1_S3_S4	0.160	100	0.220	137	0.359	224	0.376	234	0.393	245	0.378	236	0.415	258	0.377	235
C3_L1r_S1_S4	0.160	100	0.208	130	0.279	174	0.342	213	0.358	223	0.299	187	0.353	220	0.319	199
C4_L1r_S1_S3_S4	0.140	100	0.191	137	0.263	189	0.335	240	0.358	256	0.294	210	0.342	245	0.310	222
C_L1r_X1	0.100	100	0.118	117	0.165	164	0.181	180	0.207	206	0.257	256	0.204	203	0.184	183
C_X1_S1	0.130	100	0.179	138	0.271	209	0.225	173	0.253	194	0.300	231	0.294	226	0.296	228
C_S2_X1	0.170	100	0.212	124	0.296	174	0.284	167	0.317	187	0.360	212	0.336	198	0.317	187
C_L1r_X1m	0.115	100	0.148	129	0.184	160	0.216	188	0.232	202	0.291	253	-	-	0.217	188
C_S1_X1s	0.193	100	0.231	119	0.341	176	0.299	155	0.335	174	0.355	184	-	-	0.349	181
C3_L1r_S1_X1	0.105	100	0.156	149	0.194	185	0.207	198	0.224	214	0.251	240	-	-	0.215	205
<i>Backward: input prediction, graded by execution</i>																
L0	0.409	100	-	-	0.335	82	0.320	78	0.332	81	0.365	89	0.338	83	0.414	101
L1b	0.371	100	-	-	0.297	80	0.280	75	0.285	77	0.297	80	0.318	86	0.383	103
L1r	0.373	100	-	-	0.312	84	0.290	78	0.297	80	0.316	85	0.311	83	0.372	100
L2	0.376	100	-	-	0.297	79	0.284	75	0.294	78	0.319	85	0.311	83	0.364	97
S1	0.375	100	-	-	0.325	87	0.285	76	0.291	78	0.360	96	0.316	84	0.411	110
S2	0.378	100	-	-	0.345	91	0.326	86	0.325	86	0.366	97	0.366	97	0.435	115
X1	0.263	100	-	-	0.267	101	0.211	80	0.228	87	0.302	115	0.258	98	0.311	118
C_L1b_S1	0.287	100	-	-	0.253	88	0.274	95	0.258	90	0.297	103	0.306	107	0.352	123
C_L1r_S1	0.297	100	-	-	0.281	94	0.280	94	0.283	95	0.271	91	0.293	98	0.350	118
C_S1_L1r	0.303	100	-	-	0.269	89	0.274	91	0.274	91	0.290	96	0.293	97	0.356	118
C_L2_S4	0.320	100	-	-	0.275	86	0.269	84	0.286	89	0.309	96	0.313	98	0.388	121
C_L1r_S3	0.342	100	-	-	0.297	87	0.299	87	0.303	89	0.308	90	0.320	94	0.375	110
C_S4_S3	0.367	100	-	-	0.333	91	0.313	85	0.329	90	0.362	99	0.363	99	0.417	114
C3_L1r_S3_S4	0.309	100	-	-	0.271	88	0.303	98	0.300	97	0.300	97	0.317	103	0.372	121
C3_S1_S3_S4	0.353	100	-	-	0.311	88	0.289	82	0.281	80	0.340	96	0.311	88	0.408	116
C3_L1r_S1_S4	0.302	100	-	-	0.270	89	0.262	87	0.271	90	0.270	89	0.276	91	0.341	113
C4_L1r_S1_S3_S4	0.281	100	-	-	0.234	83	0.280	99	0.279	99	0.259	92	0.274	97	0.340	121
C_L1r_X1	0.218	100	-	-	0.220	101	0.193	88	0.239	109	0.265	122	0.230	105	0.275	126
C_X1_S1	0.291	100	-	-	0.283	97	0.210	72	0.220	76	0.302	104	0.269	92	0.332	114
C_S2_X1	0.323	100	-	-	0.288	89	0.229	71	0.241	75	0.306	95	0.284	88	0.350	109
C_L1r_X1m	0.232	100	-	-	0.210	91	0.217	93	0.245	106	0.258	111	0.225	97	0.274	118
C_S1_X1s	0.357	100	-	-	0.287	80	0.205	57	0.212	59	0.330	92	0.276	77	0.376	105
C3_L1r_S1_X1	0.268	100	-	-	0.234	87	0.188	70	0.194	73	0.243	91	0.241	90	0.305	114

Table 13. **Every method against every obfuscation, StarCoder2-15B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.000 [†]	-	0.008 [†]	-	0.542	-	0.529	-	0.532	-	0.513	-	0.546	-	0.539	-
L1b	0.000 [†]	-	0.000 [†]	-	0.431	-	0.487	-	0.510	-	0.402	-	0.489	-	0.426	-
L1r	0.000 [†]	-	0.000 [†]	-	0.483	-	0.485	-	0.509	-	0.448	-	0.492	-	0.476	-
L2	0.000 [†]	-	0.002 [†]	-	0.479	-	0.486	-	0.503	-	0.437	-	0.493	-	0.466	-
S1	0.000 [†]	-	0.010 [†]	-	0.508	-	0.487	-	0.501	-	0.470	-	0.510	-	0.487	-
S2	0.000 [†]	-	0.023 [†]	-	0.512	-	0.514	-	0.539	-	0.501	-	0.533	-	0.518	-
X1	0.000 [†]	-	0.009 [†]	-	0.347	-	0.320	-	0.343	-	0.381	-	0.353	-	0.343	-
C_L1b_S1	0.000 [†]	-	0.001 [†]	-	0.360	-	0.416	-	0.456	-	0.336	-	0.417	-	0.354	-
C_L1r_S1	0.000 [†]	-	0.000 [†]	-	0.393	-	0.423	-	0.440	-	0.377	-	0.419	-	0.389	-
C_S1_L1r	0.000 [†]	-	0.000 [†]	-	0.377	-	0.406	-	0.431	-	0.370	-	0.418	-	0.387	-
C_L2_S4	0.000 [†]	-	0.005 [†]	-	0.454	-	0.470	-	0.508	-	0.421	-	0.485	-	0.438	-
C_L1r_S3	0.000 [†]	-	0.000 [†]	-	0.473	-	0.483	-	0.504	-	0.435	-	0.490	-	0.459	-
C_S4_S3	0.000 [†]	-	0.028 [†]	-	0.511	-	0.516	-	0.539	-	0.490	-	0.525	-	0.506	-
C3_L1r_S3_S4	0.000 [†]	-	0.004 [†]	-	0.448	-	0.484	-	0.503	-	0.413	-	0.475	-	0.436	-
C3_S1_S3_S4	0.000 [†]	-	0.019 [†]	-	0.476	-	0.465	-	0.493	-	0.459	-	0.480	-	0.464	-
C3_L1r_S1_S4	0.000 [†]	-	0.002 [†]	-	0.370	-	0.417	-	0.439	-	0.366	-	0.413	-	0.376	-
C4_L1r_S1_S3_S4	0.000 [†]	-	0.001 [†]	-	0.357	-	0.412	-	0.437	-	0.353	-	0.401	-	0.358	-
C_L1r_X1	0.000 [†]	-	0.002 [†]	-	0.248	-	0.274	-	0.286	-	0.288	-	-	-	0.259	-
C_X1_S1	0.000 [†]	-	0.022 [†]	-	0.357	-	0.322	-	0.342	-	0.401	-	-	-	0.371	-
C_S2_X1	0.000 [†]	-	0.014 [†]	-	0.403	-	0.375	-	0.397	-	0.420	-	-	-	0.403	-
C_L1r_X1m	0.000 [†]	-	0.003 [†]	-	0.307	-	0.323	-	0.338	-	0.342	-	-	-	0.317	-
C_S1_X1s	0.000 [†]	-	0.044 [†]	-	0.434	-	0.408	-	0.434	-	0.435	-	-	-	0.427	-
C3_L1r_S1_X1	0.000 [†]	-	0.002 [†]	-	0.278	-	0.298	-	0.302	-	0.306	-	-	-	0.286	-
<i>Backward: input prediction, graded by execution</i>																
L0	0.366	100	-	-	0.411	112	0.311	85	0.399	109	0.393	107	0.435	119	0.452	124
L1b	0.216 [†]	-	-	-	0.316	-	0.301	-	0.395	-	0.265 [†]	-	0.382	-	0.364	-
L1r	0.282 [†]	-	-	-	0.354	-	0.273	-	0.369	-	0.298	-	0.368	-	0.423	-
L2	0.355	100	-	-	0.391	110	0.297	84	0.379	107	0.363	102	0.406	115	0.421	119
S1	0.235 [†]	-	-	-	0.361	-	0.257 [†]	-	0.364	-	0.342	-	0.377	-	0.421	-
S2	0.238 [†]	-	-	-	0.359	-	0.300	-	0.384	-	0.299 [†]	-	0.404	-	0.440	-
X1	0.235 [†]	-	-	-	0.254 [†]	-	0.177 [†]	-	0.281	-	0.216 [†]	-	0.261 [†]	-	0.328	-
C_L1b_S1	0.122 [†]	-	-	-	0.254 [†]	-	0.211 [†]	-	0.317	-	0.199 [†]	-	0.297	-	0.324	-
C_L1r_S1	0.165 [†]	-	-	-	0.301	-	0.214 [†]	-	0.303	-	0.257 [†]	-	0.305	-	0.388	-
C_S1_L1r	0.136 [†]	-	-	-	0.311	-	0.183 [†]	-	0.303	-	0.252 [†]	-	0.309	-	0.374	-
C_L2_S4	0.215 [†]	-	-	-	0.342	-	0.260 [†]	-	0.353	-	0.290	-	0.378	-	0.395	-
C_L1r_S3	0.243 [†]	-	-	-	0.300	-	0.282	-	0.368	-	0.248 [†]	-	0.323	-	0.383	-
C_S4_S3	0.243 [†]	-	-	-	0.368	-	0.301	-	0.386	-	0.303 [†]	-	0.389	-	0.448	-
C3_L1r_S3_S4	0.182 [†]	-	-	-	0.294	-	0.242 [†]	-	0.339	-	0.185 [†]	-	0.315	-	0.386	-
C3_S1_S3_S4	0.191 [†]	-	-	-	0.337	-	0.235 [†]	-	0.333	-	0.272 [†]	-	0.345	-	0.406	-
C3_L1r_S1_S4	0.096 [†]	-	-	-	0.293	-	0.209 [†]	-	0.270	-	0.220 [†]	-	-	-	0.353	-
C4_L1r_S1_S3_S4	0.104 [†]	-	-	-	0.289	-	0.206 [†]	-	0.273	-	0.174 [†]	-	-	-	0.358	-
C_L1r_X1	0.123 [†]	-	-	-	0.165 [†]	-	0.165 [†]	-	0.205 [†]	-	0.109 [†]	-	-	-	0.262	-
C_X1_S1	0.194 [†]	-	-	-	0.259 [†]	-	0.127 [‡]	-	0.243 [†]	-	0.181 [†]	-	-	-	0.355	-
C_S2_X1	0.184 [†]	-	-	-	0.261 [†]	-	0.196 [†]	-	0.268 [†]	-	0.192 [†]	-	-	-	0.384	-
C_L1r_X1m	0.179 [†]	-	-	-	0.191 [†]	-	0.178 [†]	-	0.234	-	0.130 [†]	-	-	-	0.267	-
C_S1_X1s	0.258 [†]	-	-	-	0.291 [†]	-	0.176 [†]	Vol. 10, No. 1, Article 1	0.180 [†]	-	0.181 [†]	-	-	-	0.384	-
C3_L1r_S1_X1	0.153 [†]	-	-	-	0.191 [†]	-	0.107 [‡]	-	0.167 [†]	-	0.059 [‡]	-	-	-	0.323	-

Table 14. **Every method against every obfuscation, Gemma-3-12B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.334†	-	0.340†	-	0.520	-	0.507	-	0.502	-	0.506	-	-	-	0.523	-
L1b	0.262†	-	0.274†	-	0.441	-	0.475	-	0.488	-	0.402	-	-	-	0.464	-
L1r	0.273†	-	0.283†	-	0.459	-	0.463	-	0.472	-	0.447	-	-	-	0.474	-
L2	0.269†	-	0.281†	-	0.456	-	0.471	-	0.478	-	0.441	-	-	-	0.465	-
S1	0.269†	-	0.289†	-	0.456	-	0.457	-	0.483	-	0.452	-	-	-	0.479	-
S2	0.290†	-	0.308†	-	0.500	-	0.503	-	0.504	-	0.489	-	-	-	0.513	-
X1	0.152	100	0.183	120	0.310	203	0.263	172	0.289	190	0.376	246	-	-	0.311	204
C_L1b_S1	0.195†	-	0.197†	-	0.333	-	0.406	-	0.420	-	0.330	-	-	-	0.359	-
C_L1r_S1	0.200†	-	0.204†	-	0.336	-	0.389	-	0.416	-	0.344	-	-	-	0.362	-
C_S1_L1r	0.181†	-	0.193†	-	0.334	-	0.393	-	0.414	-	0.329	-	-	-	0.338	-
C_L2_S4	0.249†	-	0.262†	-	0.438	-	0.460	-	0.469	-	0.424	-	-	-	0.452	-
C_L1r_S3	0.260†	-	0.274†	-	0.438	-	0.462	-	0.469	-	0.423	-	-	-	0.450	-
C_S4_S3	0.297†	-	0.308†	-	0.491	-	0.506	-	0.503	-	0.481	-	-	-	0.496	-
C3_L1r_S3_S4	0.232†	-	0.256†	-	0.419	-	0.454	-	0.467	-	0.417	-	-	-	0.440	-
C3_S1_S3_S4	0.220†	-	0.256†	-	0.425	-	0.452	-	0.465	-	0.431	-	-	-	0.449	-
C3_L1r_S1_S4	0.176†	-	0.191†	-	0.320	-	0.390	-	0.415	-	0.330	-	-	-	0.341	-
C4_L1r_S1_S3_S4	0.170†	-	0.172†	-	0.302	-	0.394	-	0.413	-	0.323	-	-	-	0.332	-
C_L1r_X1	0.102	100	0.126	123	0.195	191	0.206	202	0.224	219	0.285	279	-	-	0.203	198
C_X1_S1	0.172	100	0.193	113	0.317	185	0.271	158	0.296	173	0.358	208	-	-	0.312	182
C_S2_X1	0.201	100	0.222	110	0.368	183	0.349	173	0.364	181	0.412	205	-	-	0.368	183
C_L1r_X1m	0.108†	-	0.115†	-	0.220	-	0.235	-	0.244	-	0.312	-	-	-	0.231	-
C_S1_X1s	0.255	100	0.275	108	0.395	155	0.374	147	0.395	155	0.401	158	-	-	0.393	154
C3_L1r_S1_X1	0.136	100	0.147†	-	0.225	165	0.236	174	0.258	189	0.260	191	-	-	0.246	181
<i>Backward: input prediction, graded by execution</i>																
L0	0.453	100	-	-	0.440	97	0.447	99	0.355	78	0.469	104	-	-	0.481	106
L1b	0.405	100	-	-	0.353	87	0.427	106	0.309†	-	0.395	98	-	-	0.425	105
L1r	0.420	100	-	-	0.394	94	0.411	98	0.280†	-	0.428	102	-	-	0.440	105
L2	0.403	100	-	-	0.378	94	0.405	100	0.295†	-	0.431	107	-	-	0.420	104
S1	0.424	100	-	-	0.375	88	0.421	99	0.256†	-	0.461	109	-	-	0.435	103
S2	0.443	100	-	-	0.444	100	0.457	103	0.277†	-	0.454	102	-	-	0.484	109
X1	0.333	100	-	-	0.305	91	0.333	100	0.199†	-	0.393	118	-	-	0.371	111
C_L1b_S1	0.349	100	-	-	0.317	91	0.366	105	0.214†	-	0.370	106	-	-	0.396	113
C_L1r_S1	0.356	100	-	-	0.319	89	0.379	106	0.203†	-	0.397	111	-	-	0.386	108
C_S1_L1r	0.337	100	-	-	0.319	95	0.379	112	0.207†	-	0.407	121	-	-	0.405	120
C_L2_S4	0.387	100	-	-	0.381	98	0.416	108	0.222†	-	0.427	110	-	-	0.430	111
C_L1r_S3	0.427	100	-	-	0.402	94	0.423	99	0.254†	-	0.438	103	-	-	0.443	104
C_S4_S3	0.434	100	-	-	0.438	101	0.462	106	0.274†	-	0.469	108	-	-	0.472	109
C3_L1r_S3_S4	0.398	100	-	-	0.386	97	0.420	106	0.223†	-	0.447	112	-	-	0.441	111
C3_S1_S3_S4	0.387	100	-	-	0.354	92	0.419	108	0.240†	-	0.440	114	-	-	0.430	111
C3_L1r_S1_S4	0.347	100	-	-	0.302	87	0.376	109	0.181†	-	0.377	109	-	-	0.377	109
C4_L1r_S1_S3_S4	0.329	100	-	-	0.293	89	0.364	111	0.180†	-	0.399	121	-	-	0.370	112
C_L1r_X1	0.258	100	-	-	0.242	94	0.279	108	0.121†	-	0.329	128	-	-	0.274	106
C_X1_S1	0.348	100	-	-	0.282	81	0.331	95	0.132†	-	0.410	118	-	-	0.375	108
C_S2_X1	0.360	100	-	-	0.339	94	0.357	99	0.210†	-	0.398	111	-	-	0.395	110
C_L1r_X1m	0.273	100	-	-	0.269	99	0.323	118	0.146†	-	0.349	128	-	-	0.285	105
C_S1_X1s	0.412	100	-	-	0.371	90	0.383	93	0.178†	-	0.448	109	-	-	0.438	106
C3_L1r_S1_X1	-	-	-	-	-	81	0.270	97	0.115†	-	0.337	121	-	-	0.300	107

Table 15. **Every method against every obfuscation, CodeGemma-7B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.237	100	0.267†	-	0.465	197	0.439	186	0.445	188	0.462	195	0.474	200	0.465	196
L1b	0.185	100	0.209†	-	0.382	207	0.425	230	0.435	236	0.345	187	0.443	240	0.391	212
L1r	0.213	100	0.225†	-	0.399	187	0.417	196	0.417	196	0.390	183	0.426	200	0.411	193
L2	0.208	100	0.230†	-	0.412	198	0.422	203	0.422	203	0.392	188	0.440	211	0.406	195
S1	0.195	100	0.212	109	0.405	208	0.415	213	0.421	216	0.423	217	0.457	235	0.414	212
S2	0.198	100	0.242†	-	0.450	227	0.429	217	0.448	226	0.444	224	0.479	242	0.448	226
X1	0.143	100	0.157	110	0.271	189	0.251	175	0.247	172	0.348	243	-	-	0.274	191
C_L1b_S1	0.148	100	0.163	110	0.286	192	0.372	251	0.368	248	0.287	194	-	-	0.302	204
C_L1r_S1	0.144	100	0.144	100	0.315	219	0.373	259	0.367	256	0.304	212	-	-	0.318	222
C_S1_L1r	0.132	100	0.149	113	0.288	217	0.363	274	0.361	272	0.299	226	-	-	0.300	227
C_L2_S4	0.176	100	0.208†	-	0.391	222	0.423	240	0.421	239	0.374	212	-	-	0.391	222
C_L1r_S3	0.183	100	0.216†	-	0.396	216	0.415	226	0.410	224	0.375	205	-	-	0.393	214
C_S4_S3	0.202	100	0.235†	-	0.448	222	0.430	213	0.441	219	0.446	221	-	-	0.445	221
C3_L1r_S3_S4	0.157	100	0.182†	-	0.376	239	0.407	259	0.411	261	0.361	229	-	-	0.372	237
C3_S1_S3_S4	0.158	100	0.185	117	0.367	233	0.395	250	0.405	256	0.392	248	-	-	0.378	239
C3_L1r_S1_S4	0.126	100	0.137	109	0.289	229	0.367	291	0.366	290	0.301	239	-	-	0.303	241
C4_L1r_S1_S3_S4	0.124	100	0.125†	-	0.271	219	0.361	292	0.359	290	0.285	230	-	-	0.303	245
C_L1r_X1	0.118	100	0.110	93	0.194	164	0.212	180	0.197	167	0.245	208	-	-	0.184	156
C_X1_S1	0.144	100	0.152	105	0.285	197	0.261	181	0.247	171	0.339	235	-	-	0.288	199
C_S2_X1	0.188	100	0.198	105	0.344	183	0.323	172	0.326	173	0.379	201	-	-	0.340	181
C_L1r_X1m	0.130	100	0.112	86	0.214	164	0.256	196	0.234	180	0.288	221	-	-	0.201	154
C_S1_X1s	0.186	100	0.210	113	0.338	182	0.351	189	0.339	182	0.386	208	-	-	0.358	193
C3_L1r_S1_X1	0.134	100	0.118	88	0.218	163	0.241	179	0.246	183	0.254	189	-	-	0.224	167
<i>Backward: input prediction, graded by execution</i>																
L0	0.266	100	-	-	0.364	137	0.324	122	0.098‡	-	0.256	96	0.323	121	0.329	124
L1b	0.268	100	-	-	0.309	115	0.298	111	0.067‡	-	0.193	72	0.307	115	0.301	112
L1r	0.251	100	-	-	0.333	133	0.301	120	0.107‡	-	0.225	90	0.294	117	0.303	121
L2	0.239	100	-	-	0.327	136	0.291	122	0.078‡	-	0.220	92	0.287	120	0.299	125
S1	0.234	100	-	-	0.357	153	0.337	144	0.125‡	-	0.231	99	0.320	137	0.312	134
S2	0.270	100	-	-	0.372	138	0.351	130	0.126‡	-	0.248	92	0.359	133	0.330	122
X1	0.190	100	-	-	0.260	137	0.217	114	0.049‡	-	0.235	123	-	-	0.264	139
C_L1b_S1	0.229	100	-	-	0.274	120	0.276	121	0.089‡	-	0.176†	-	-	-	0.297	130
C_L1r_S1	0.170	100	-	-	0.289	170	0.290	171	0.086‡	-	0.208	123	-	-	0.271	160
C_S1_L1r	0.168	100	-	-	0.284	169	0.316	188	0.099‡	-	0.203	120	-	-	0.281	167
C_L2_S4	0.239	100	-	-	0.318	133	0.299	125	0.070‡	-	0.238	100	-	-	0.295	124
C_L1r_S3	0.223	100	-	-	0.326	147	0.314	141	0.150†	-	0.223	100	-	-	0.293	131
C_S4_S3	0.257	100	-	-	0.372	145	0.357	139	0.130‡	-	0.242	94	-	-	0.348	136
C3_L1r_S3_S4	0.187	100	-	-	0.318	170	0.312	167	0.083‡	-	0.221	118	-	-	0.294	157
C3_S1_S3_S4	0.231	100	-	-	0.338	146	0.332	143	0.138‡	-	0.234	101	-	-	0.306	132
C3_L1r_S1_S4	0.173	100	-	-	0.274	158	0.286	165	0.064‡	-	0.191	110	-	-	0.267	154
C4_L1r_S1_S3_S4	0.150	100	-	-	0.267	178	0.290	194	0.081‡	-	0.177	118	-	-	0.270	180
C_L1r_X1	0.133†	-	-	-	0.208	-	0.174	-	0.036‡	-	0.178	-	-	-	0.202	-
C_X1_S1	0.202	100	-	-	0.303	150	0.209	103	0.034‡	-	0.227	112	-	-	0.280	138
C_S2_X1	0.227	100	-	-	0.300	132	0.241	106	0.031‡	-	0.210†	-	-	-	0.293	129
C_L1r_X1m	0.146†	-	-	-	0.231	-	0.211	-	0.049‡	-	0.205	-	-	-	0.230	-
C_S1_X1s	0.230	100	-	-	0.322	140	0.230	100	0.037‡	-	0.243	106	-	-	0.302	132
C3_L1r_S1_X1	0.168	100	-	-	0.230	137	0.140†	-	0.013‡	-	0.191	114	-	-	0.232	138

Table 16. **Every method against every obfuscation, Granite-3.1-8B.** Raw accuracy (strict exact match), with % *b* after each system: its accuracy as a percentage of base’s on the same condition. *base* is the untuned model; *ICL* its one-shot prompt; *clean LoRA* is tuned on unobfuscated code; *breadth* on all five seen transforms; *anchored* is the paired-consistency objective; *family* is trained on the unseen family’s sibling; *mixture* is the learned-gate mixture of per-transform specialists; *merge* is the DARE-TIES merge of them (keys in order: base, base_1shot, tuned_L0, mono_all, cons_lam3, tuned_X1, mole_router, merge_dare_ties). **Backward is graded by execution:** the produced call is run against the program as shown and any call that returns the gold value is correct, because inversion is many-to-one — 21.5 % of the ladder’s gold return values have many valid inputs, so requiring the one recorded input would mark correct answers wrong. The stricter exact-argument rate is a secondary column in the repository’s master tables. †: format-failure rate above 0.25, mostly replies that are not call-shaped. ‡: format-failure rate above 0.25, but most of those replies are exactly the gold *forward* answer — the arm answered the other question rather than failing to parse. ‘-’: not run.

Condition	base		ICL		clean LoRA		breadth		anchored		family		mixture		merge	
	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b	acc	% b
<i>Forward: output prediction</i>																
L0	0.285	100	0.293	103	0.430	151	0.416	146	0.396	139	0.413	145	0.438	154	0.433	152
L1b	0.267	100	0.267	100	0.338	127	0.391	147	0.381	143	0.321	120	0.396	149	0.350	131
L1r	0.268	100	0.260	97	0.375	140	0.383	143	0.378	141	0.376	140	0.391	146	0.384	144
L2	0.251	100	0.248	99	0.372	148	0.387	154	0.377	150	0.366	145	0.400	159	0.378	150
S1	0.254	100	0.253	99	0.346	136	0.379	149	0.358	141	0.350	138	0.403	158	0.371	146
S2	0.272	100	0.276	101	0.386	142	0.415	152	0.394	144	0.398	146	0.422	155	0.402	148
X1	0.147	100	0.160	109	0.216	147	0.207	141	0.188	128	0.282	192	0.244	166	0.231	157
C_L1b_S1	0.219	100	0.191	87	0.263	120	0.342	157	0.337	154	0.261	119	0.324	148	0.279	127
C_L1r_S1	0.204	100	0.187	91	0.279	137	0.344	168	0.338	165	0.291	143	0.334	164	0.301	147
C_S1_L1r	0.224	100	0.187	83	0.279	124	0.326	146	0.319	142	0.291	130	0.327	146	0.293	131
C_L2_S4	0.242	100	0.245	101	0.334	138	0.376	156	0.376	156	0.341	141	0.384	159	0.351	145
C_L1r_S3	0.259	100	0.259	100	0.350	135	0.382	148	0.376	145	0.359	139	0.378	146	0.365	141
C_S4_S3	0.270	100	0.282	104	0.380	141	0.412	152	0.389	144	0.383	142	0.429	159	0.394	146
C3_L1r_S3_S4	0.257	100	0.245	96	0.333	130	0.380	148	0.375	146	0.340	132	0.375	146	0.348	136
C3_S1_S3_S4	0.242	100	0.248	103	0.316	131	0.372	154	0.358	148	0.330	137	0.380	157	0.342	142
C3_L1r_S1_S4	0.221	100	0.200	90	0.279	126	0.333	151	0.330	149	0.283	128	-	-	0.287	130
C4_L1r_S1_S3_S4	0.203	100	0.205	101	0.258	127	0.325	160	0.334	165	0.270	133	-	-	0.281	139
C_L1r_X1	0.136	100	0.124	92	0.157	115	0.172	127	0.163	120	0.245	181	-	-	0.173	127
C_X1_S1	0.153	100	0.169	111	0.222	146	0.203	133	0.186	122	0.272	178	-	-	0.244	160
C_S2_X1	0.209	100	0.217	104	0.263	126	0.252	121	0.240	115	0.323	155	-	-	0.272	130
C_L1r_X1m	0.158	100	0.145	92	0.188	119	0.204	130	0.183	116	0.279	177	-	-	0.211	134
C_S1_X1s	0.206	100	0.215	104	0.298	145	0.267	129	0.269	131	0.318	154	-	-	0.293	142
C3_L1r_S1_X1	0.167	100	0.147	88	0.186	111	0.195	117	0.198	119	0.238	142	-	-	0.217	130
<i>Backward: input prediction, graded by execution</i>																
L0	0.291	100	-	-	0.206 [†]	-	0.261	90	0.117 [†]	-	0.272	94	0.268	92	0.304	105
L1b	0.255	100	-	-	0.156 [†]	-	0.228	90	0.117 [†]	-	0.211	83	0.252	99	0.278	109
L1r	0.284	100	-	-	0.209 [†]	-	0.268	94	0.150 [†]	53	0.262	92	0.280	99	0.299	105
L2	0.247	100	-	-	0.209 [†]	-	0.236	96	0.133 [†]	-	0.222	90	0.244	99	0.294	119
S1	0.305	100	-	-	0.196 [†]	-	0.220	72	0.099 [†]	-	0.236	77	0.260	85	0.284	93
S2	0.329	100	-	-	0.182 [†]	-	0.244	74	0.098 [†]	-	0.266	81	0.278	84	0.315	96
X1	0.198 [†]	-	-	-	0.123 [†]	-	0.156 [†]	-	0.048 [‡]	-	0.207	-	0.210	-	0.243	-
C_L1b_S1	0.216	100	-	-	0.136 [†]	-	0.199	92	0.082 [†]	-	0.194	90	0.234	108	0.228	106
C_L1r_S1	0.284	100	-	-	0.206	72	0.225	79	0.126 [†]	-	0.234	82	0.250	88	0.267	94
C_S1_L1r	0.262	100	-	-	0.208	79	0.241	92	0.132 [†]	-	0.242	92	0.254	97	0.276	105
C_L2_S4	0.278	100	-	-	0.182 [†]	-	0.225	81	0.117 [†]	-	0.213	77	0.242	87	0.287	103
C_L1r_S3	0.296	100	-	-	0.160 [†]	-	0.240	81	0.145 [†]	-	0.251	85	0.279	94	0.296	100
C_S4_S3	0.334	100	-	-	0.180 [†]	-	0.254	76	0.100 [†]	-	0.254	76	0.290	87	0.313	94
C3_L1r_S3_S4	0.331	100	-	-	0.180 [†]	-	0.238	72	0.135 [†]	-	0.235	71	0.279	84	0.276	83
C3_S1_S3_S4	0.323	100	-	-	0.181 [†]	-	0.214	66	0.062 [†]	-	0.242	75	0.273	84	0.289	89
C3_L1r_S1_S4	0.293	100	-	-	0.192	66	0.223	76	0.111 [†]	-	0.238	81	0.246	84	0.262	90
C4_L1r_S1_S3_S4	0.305	100	-	-	0.175 [†]	-	0.216	71	0.106 [†]	-	0.239	79	0.253	83	0.272	89
C_L1r_X1	0.165 [†]	-	-	-	0.102 [†]	-	0.163	-	0.072 [†]	-	0.190	-	0.170	-	0.201	-
C_X1_S1	0.234	100	-	-	0.156 [†]	-	0.161	69	0.034 [‡]	-	0.242	104	0.228	98	0.251	108
C_S2_X1	0.300	100	-	-	0.151 [†]	-	0.150 [†]	-	0.043 [‡]	-	0.235	78	-	-	0.277	92
C_L1r_X1m	0.159 [†]	-	-	-	0.117 [†]	-	0.180	-	0.091 [†]	-	0.188	-	-	-	0.196	-
C_S1_X1s	0.282	100	-	-	0.141 [†]	-	0.173	61	0.037 [‡]	-	0.234	83	-	-	0.268	95
C3_L1r_S1_X1	0.259	100	-	-	0.163 [†]	-	0.156	60	0.057 [‡]	-	0.226	87	-	-	0.239	92

Table 17. **Master grid, CodeLlama-7B.** Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.257	100	0.302	+4.5	117	0.427	+17.0	166	0.413	+15.6	161	0.422	+16.5	164	0.413	+15.6	161
L1b	0.198	77	0.255	+5.7	99	0.358	+16.0	139	0.388	+19.0	151	0.402	+20.4	156	0.346	+14.8	135
L1r	0.207	81	0.265	+5.8	103	0.378	+17.1	147	0.387	+18.0	151	0.395	+18.8	154	0.360	+15.3	140
L2	0.202	79	0.272	+7.0	106	0.382	+18.0	149	0.378	+17.7	147	0.398	+19.6	155	0.354	+15.2	138
S1	0.168	66	0.216	+4.7	84	0.379	+21.0	147	0.389	+22.1	151	0.400	+23.2	156	0.370	+20.2	144
S2	0.193	75	fmt			0.389	+19.6	152	0.403	+20.9	157	0.415	+22.2	162	0.383	+19.0	149
X1	0.119	46	-	-	-	0.269	+14.9	105	0.231	+11.2	90	0.286	+16.6	111	0.316	+19.7	123
C_L1b_S1	0.141	55	-	-	-	0.279	+13.7	108	0.345	+20.4	134	0.360	+21.9	140	0.278	+13.6	108
C_L1r_S1	0.131	51	-	-	-	0.294	+16.3	114	0.350	+21.9	136	0.356	+22.5	139	0.296	+16.5	115
C_S1_L1r	0.135	53	-	-	-	0.285	+15.0	111	0.326	+19.2	127	0.349	+21.4	136	0.297	+16.2	116
C_L2_S4	0.176	68	-	-	-	0.355	+17.9	138	0.385	+20.9	150	0.392	+21.7	153	0.343	+16.7	134
C_L1r_S3	0.172	67	-	-	-	0.360	+18.8	140	0.380	+20.8	148	0.387	+21.5	151	0.350	+17.8	136
C_S4_S3	0.186	72	-	-	-	0.392	+20.6	153	0.403	+21.7	157	0.422	+23.6	164	0.382	+19.6	149
seen stacks, pooled (6)	0.157	61	-	-	-	0.327	+17.1	127	0.365	+20.8	142	0.378	+22.1	147	0.324	+16.8	126
C3_L1r_S3_S4	0.150	58	-	-	-	0.340	+19.0	132	0.377	+22.7	147	0.380	+23.0	148	0.343	+19.3	133
C3_S1_S3_S4	0.128	50	-	-	-	0.354	+22.6	138	0.369	+24.1	144	0.385	+25.7	150	0.343	+21.5	133
C3_L1r_S1_S4	0.126	49	-	-	-	0.289	+16.3	112	0.353	+22.7	137	0.354	+22.7	138	0.303	+17.7	118
C4_L1r_S1_S3_S4	0.089	35	-	-	-	0.278	+18.8	108	0.336	+24.7	131	0.356	+26.7	139	0.292	+20.3	114
depth-3/4 stacks, pooled (4)	0.123	48	-	-	-	0.315	+19.2	123	0.359	+23.5	140	0.369	+24.5	144	0.320	+19.7	125
C_L1r_X1	0.102	40	-	-	-	0.184	+8.2	72	0.199	+9.6	77	0.222	+12.0	87	0.254	+15.2	99
C_X1_S1	0.104	40	-	-	-	0.274	+17.0	107	0.230	+12.6	89	0.267	+16.3	104	0.326	+22.2	127
C_S2_X1	0.145	57	-	-	-	0.313	+16.8	122	0.285	+13.9	111	0.332	+18.6	129	0.347	+20.2	135
unseen-containing stacks, pooled (3)	0.117	46	-	-	-	0.257	+14.0	100	0.238	+12.1	93	0.274	+15.7	107	0.309	+19.2	120
C_L1r_X1m	0.112	44	-	-	-	0.220	+10.7	85	0.232	+12.0	90	0.249	+13.7	97	0.281	+16.9	110
C_S1_X1s	0.151	59	-	-	-	0.334	+18.3	130	0.302	+15.1	117	0.355	+20.4	138	0.341	+19.0	133
C3_L1r_S1_X1	0.104	40	-	-	-	0.219	+11.5	85	0.222	+11.8	86	0.250	+14.6	97	0.271	+16.7	106
<i>Backward: input prediction, graded by execution</i>																	
L0	0.285	100	-	-	-	0.324	+3.9	114	0.276	-0.8	97	0.297	+1.3	104	0.312	+2.7	109
L1b	0.265	93	-	-	-	0.251	-1.3	88	0.279	+1.4	98	0.294	+2.9	103	0.265	+0.1	93
L1r	0.315	111	-	-	-	0.327	+1.1	115	0.276	-3.9	97	0.293	-2.3	103	0.327	+1.1	115
L2	0.282	99	-	-	-	0.324	+4.2	114	0.276	-0.6	97	0.275	-0.7	97	0.324	+4.2	114
S1	0.261	92	-	-	-	fmt			0.262	+0.1	92	0.280	+1.9	98	0.280	+1.8	98
S2	0.330	116	-	-	-	fmt			0.294	-3.6	103	0.332	+0.2	117	0.330	+0.0	116
X1	0.257	90	-	-	-	fmt			0.237	-2.0	83	0.281	+2.5	99	0.291	+3.4	102
C_L1b_S1	0.218	76	-	-	-	fmt			0.242	+2.5	85	0.270	+5.3	95	0.246	+2.8	86
C_L1r_S1	0.301	106	-	-	-	fmt			0.255	-4.6	90	0.270	-3.1	95	0.300	-0.2	105
C_S1_L1r	0.282	99	-	-	-	0.205	-7.7	72	0.267	-1.5	94	0.285	+0.2	100	0.262	-2.0	92
C_L2_S4	0.291	102	-	-	-	fmt			0.260	-3.1	91	0.283	-0.8	99	0.315	+2.4	111
C_L1r_S3	0.296	104	-	-	-	0.267	-2.9	94	0.278	-1.9	97	0.315	+1.9	111	0.330	+3.4	116
C_S4_S3	0.315	111	-	-	-	fmt			0.282	-3.4	99	0.333	+1.7	117	0.324	+0.8	114
seen stacks, pooled (6)	0.284	100	-	-	-	fmt			0.264	-2.0	93	0.293	+0.9	103	0.296	+1.2	104
C3_L1r_S3_S4	0.303	106	-	-	-	fmt			0.268	-3.5	94	0.302	-0.1	106	0.306	+0.4	108
C3_S1_S3_S4	0.292	103	-	-	-	fmt			0.234	-5.8	82	0.313	+2.1	110	0.291	-0.1	102
C3_L1r_S1_S4	0.299	105	-	-	-	fmt			0.229	-7.0	80	0.278	-2.2	97	0.262	-3.7	92
C4_L1r_S1_S3_S4	0.282	99	-	-	-	fmt			0.217	-6.5	76	0.288	+0.6	101	0.273	-1.0	96
depth-3/4 stacks, pooled (4)	0.294	103	-	-	-	fmt			0.237	-5.7	83	0.295	+0.1	104	0.283	-1.1	99
C_L1r_X1	0.253	89	-	-	-	0.210	-4.3	74	0.200	-5.3	70	0.250	-0.2	88	0.245	-0.7	86
C_X1_S1	0.251	88	-	-	-	fmt			0.201	-5.0	71	0.276	+2.5	97	0.255	+0.4	90
C_S2_X1	0.311	109	-	-	-	fmt			0.217	-9.4	76	0.305	-0.5	107	0.280	-3.0	98
unseen-containing stacks, pooled (3)	0.271	95	-	-	-	fmt			0.206	-6.5	72	0.277	+0.6	97	0.260	-1.1	91
C_L1r_X1m	0.250	88	-	-	-	0.220	-2.9	77	0.223	-2.7	78	0.250	+0.0	88	0.233	-1.7	82
C_S1_X1s	0.277	97	-	-	-	fmt			0.195	-8.1	69	0.288	+1.1	101	0.270	-0.6	95
C3_L1r_S1_X1	0.263	92	-	-	-	fmt			fmt			0.231	-3.2	81	0.254	-0.9	89

Table 18. **Master grid, CodeLlama-13B.** Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.252	100	fmt	fmt	fmt	0.469	+21.7	186	0.446	+19.3	177	0.462	+21.0	183	0.463	+21.1	184
L1b	0.222	88	fmt	fmt	fmt	0.397	+17.6	158	0.423	+20.1	168	0.438	+21.7	174	0.379	+15.7	150
L1r	0.225	89	fmt	fmt	fmt	0.415	+19.0	165	0.420	+19.5	167	0.439	+21.4	174	0.407	+18.1	161
L2	0.228	90	fmt	fmt	fmt	0.407	+17.8	161	0.417	+18.9	166	0.432	+20.4	171	0.401	+17.3	159
S1	0.213	85	0.216	+0.2	86	0.408	+19.5	162	0.413	+20.0	164	0.435	+22.1	172	0.410	+19.6	163
S2	0.184	73	fmt	fmt	fmt	0.422	+23.8	168	0.445	+26.0	176	0.469	+28.4	186	0.422	+23.8	168
X1	0.149	59	-	-	-	0.294	+14.5	117	0.254	+10.5	101	0.292	+14.3	116	0.353	+20.4	140
C_L1b_S1	0.183	72	-	-	-	0.318	+13.6	126	0.367	+18.4	146	0.375	+19.2	149	0.320	+13.7	127
C_L1r_S1	0.175	69	-	-	-	0.318	+14.4	126	0.371	+19.6	147	0.376	+20.1	149	0.342	+16.8	136
C_S1_L1r	0.172	68	-	-	-	0.315	+14.4	125	0.359	+18.8	142	0.363	+19.2	144	0.350	+17.8	139
C_L2_S4	0.200	79	-	-	-	0.389	+19.0	154	0.422	+22.2	167	0.432	+23.2	171	0.385	+18.5	153
C_L1r_S3	0.186	74	-	-	-	0.407	+22.0	161	0.409	+22.3	162	0.434	+24.8	172	0.399	+21.3	158
C_S4_S3	0.192	76	-	-	-	0.424	+23.2	168	0.436	+24.4	173	0.459	+26.7	182	0.433	+24.1	172
seen stacks, pooled (6)	0.185	73	-	-	-	0.362	+17.8	144	0.394	+21.0	156	0.407	+22.2	161	0.372	+18.7	147
C3_L1r_S3_S4	0.166	66	-	-	-	0.382	+21.6	152	0.410	+24.4	163	0.429	+26.3	170	0.382	+21.5	151
C3_S1_S3_S4	0.147	58	-	-	-	0.367	+22.0	146	0.408	+26.1	162	0.423	+27.6	168	0.384	+23.7	152
C3_L1r_S1_S4	0.163	65	-	-	-	0.314	+15.1	124	0.367	+20.4	146	0.367	+20.4	146	0.349	+18.6	138
C4_L1r_S1_S3_S4	0.117	46	-	-	-	0.300	+18.4	119	0.353	+23.6	140	0.360	+24.3	143	0.331	+21.5	131
depth-3/4 stacks, pooled (4)	0.148	59	-	-	-	0.341	+19.3	135	0.384	+23.6	152	0.395	+24.7	157	0.361	+21.3	143
C_L1r_X1	0.123	49	-	-	-	0.199	+7.7	79	0.213	+9.0	84	0.236	+11.3	93	0.284	+16.1	113
C_X1_S1	0.138	55	-	-	-	0.295	+15.7	117	0.270	+13.2	107	0.307	+16.8	122	0.341	+20.3	135
C_S2_X1	0.160	64	-	-	-	0.335	+17.4	133	0.334	+17.4	133	0.356	+19.6	141	0.376	+21.6	149
unseen-containing stacks, pooled (3)	0.140	56	-	-	-	0.276	+13.6	110	0.272	+13.2	108	0.300	+15.9	119	0.334	+19.3	132
C_L1r_X1m	0.138	55	-	-	-	0.246	+10.8	98	0.248	+11.0	98	0.279	+14.1	110	0.319	+18.2	127
C_S1_X1s	0.179	71	-	-	-	0.364	+18.5	144	0.345	+16.6	137	0.360	+18.1	143	0.368	+18.9	146
C3_L1r_S1_X1	0.128	51	-	-	-	0.233	+10.5	92	0.234	+10.6	93	0.249	+12.1	99	0.291	+16.3	116
<i>Backward: input prediction, graded by execution</i>																	
L0	0.285	100	-	-	-	0.269	-1.6	94	fmt	fmt	fmt	0.231	-5.4	81	fmt	fmt	fmt
L1b	0.277	97	-	-	-	0.285	+0.8	100	fmt	fmt	fmt	0.253	-2.5	89	fmt	fmt	fmt
L1r	0.327	115	-	-	-	0.347	+2.0	122	fmt	fmt	fmt	0.229	-9.8	80	fmt	fmt	fmt
L2	0.279	98	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	0.206	-7.2	72	fmt	fmt	fmt
S1	0.304	107	-	-	-	0.341	+3.7	120	fmt	fmt	fmt	0.199	-10.6	70	fmt	fmt	fmt
S2	0.293	103	-	-	-	0.317	+2.4	111	fmt	fmt	fmt	0.238	-5.5	84	fmt	fmt	fmt
X1	0.258	91	-	-	-	0.274	+1.6	96	fmt	fmt	fmt	0.219	-4.0	77	fmt	fmt	fmt
C_L1b_S1	0.253	89	-	-	-	0.326	+7.3	114	fmt	fmt	fmt	0.201	-5.2	71	fmt	fmt	fmt
C_L1r_S1	0.325	114	-	-	-	0.360	+3.5	126	fmt	fmt	fmt	0.203	-12.1	71	fmt	fmt	fmt
C_S1_L1r	0.323	113	-	-	-	0.351	+2.8	123	fmt	fmt	fmt	0.230	-9.3	81	fmt	fmt	fmt
C_L2_S4	0.274	96	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_S3	0.305	107	-	-	-	0.361	+5.6	127	fmt	fmt	fmt	0.220	-8.5	77	fmt	fmt	fmt
C_S4_S3	0.293	103	-	-	-	0.326	+3.4	114	fmt	fmt	fmt	0.239	-5.3	84	fmt	fmt	fmt
seen stacks, pooled (6)	0.295	104	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S3_S4	0.310	109	-	-	-	0.337	+2.7	118	fmt	fmt	fmt	0.224	-8.6	79	fmt	fmt	fmt
C3_S1_S3_S4	0.295	103	-	-	-	0.329	+3.5	116	fmt	fmt	fmt	0.235	-5.9	83	fmt	fmt	fmt
C3_L1r_S1_S4	0.316	111	-	-	-	0.353	+3.7	124	fmt	fmt	fmt	0.211	-10.5	74	fmt	fmt	fmt
C4_L1r_S1_S3_S4	0.310	109	-	-	-	0.333	+2.3	117	fmt	fmt	fmt	0.222	-8.9	78	fmt	fmt	fmt
depth-3/4 stacks, pooled (4)	0.308	108	-	-	-	0.338	+3.1	119	fmt	fmt	fmt	0.223	-8.5	78	fmt	fmt	fmt
C_L1r_X1	0.281	99	-	-	-	0.254	-2.6	89	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_X1_S1	0.289	101	-	-	-	0.264	-2.5	93	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_S2_X1	0.271	95	-	-	-	0.279	+0.8	98	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
unseen-containing stacks, pooled (3)	0.280	98	-	-	-	0.266	-1.4	93	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_X1m	0.291	102	-	-	-	0.259	-3.1	91	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C_S1_X1s	0.301	106	-	-	-	0.290	-1.1	102	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S1_X1	0.304	107	-	-	-	0.293	-1.0	103	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt

Table 19. **Master grid, CodeLlama-34B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base	acc	base_1shot	acc	Δ	tuned_L0	acc	Δ	mono_all	acc	Δ	cons_lam3	acc	Δ	tuned_X1	acc	Δ
<i>Forward: output prediction</i>																	
L0	fmt	fmt	fmt	fmt	fmt	0.520	fmt	fmt	0.494	fmt	fmt	0.525	fmt	fmt	0.502	fmt	fmt
L1b	0.205	fmt	fmt	fmt	fmt	0.417	+21.2	fmt	0.469	+26.4	fmt	0.494	+28.9	fmt	0.401	+19.6	fmt
L1r	0.220	fmt	fmt	fmt	fmt	0.469	+25.0	fmt	0.457	+23.8	fmt	0.487	+26.8	fmt	0.459	+24.0	fmt
L2	0.231	fmt	fmt	fmt	fmt	0.466	+23.5	fmt	0.465	+23.5	fmt	0.478	+24.7	fmt	0.456	+22.6	fmt
S1	0.222	fmt	fmt	fmt	fmt	0.472	+24.9	fmt	0.467	+24.5	fmt	0.496	+27.3	fmt	0.461	+23.9	fmt
S2	0.191	fmt	fmt	fmt	fmt	0.484	+29.3	fmt	0.484	+29.3	fmt	0.525	+33.5	fmt	0.478	+28.7	fmt
X1	0.135	fmt	-	-	-	0.325	+19.0	fmt	0.298	+16.3	fmt	0.334	+19.9	fmt	0.381	+24.5	fmt
C_L1b_S1	0.156	fmt	-	-	-	0.322	+16.6	fmt	0.399	+24.3	fmt	0.421	+26.5	fmt	0.336	+18.0	fmt
C_L1r_S1	0.184	fmt	-	-	-	0.351	+16.8	fmt	0.393	+21.0	fmt	0.427	+24.3	fmt	0.366	+18.2	fmt
C_S1_L1r	0.180	fmt	-	-	-	0.367	+18.8	fmt	0.405	+22.5	fmt	0.425	+24.6	fmt	0.379	+20.0	fmt
C_L2_S4	0.191	fmt	-	-	-	0.431	+24.1	fmt	0.454	+26.3	fmt	0.478	+28.7	fmt	0.431	+24.1	fmt
C_L1r_S3	0.169	fmt	-	-	-	0.451	+28.1	fmt	0.454	+28.5	fmt	0.482	+31.3	fmt	0.437	+26.8	fmt
C_S4_S3	0.199	fmt	-	-	-	0.482	+28.3	fmt	0.487	+28.7	fmt	0.521	+32.2	fmt	0.476	+27.7	fmt
<i>seen stacks, pooled (6)</i>	0.180	fmt	-	-	-	0.401	+22.1	fmt	0.432	+25.2	fmt	0.459	+27.9	fmt	0.404	+22.4	fmt
C3_L1r_S3_S4	0.148	fmt	-	-	-	0.416	+26.8	fmt	0.452	+30.4	fmt	0.473	+32.5	fmt	0.424	+27.5	fmt
C3_S1_S3_S4	0.128	fmt	-	-	-	0.428	+30.1	fmt	0.454	+32.7	fmt	0.472	+34.5	fmt	0.440	+31.2	fmt
C3_L1r_S1_S4	0.147	fmt	-	-	-	0.338	+19.2	fmt	0.390	+24.3	fmt	0.412	+26.5	fmt	0.359	+21.2	fmt
C4_L1r_S1_S3_S4	0.105	fmt	-	-	-	0.316	+21.1	fmt	0.372	+26.7	fmt	0.413	+30.8	fmt	0.354	+24.9	fmt
<i>depth-3/4 stacks, pooled (4)</i>	0.132	fmt	-	-	-	0.375	+24.3	fmt	0.417	+28.5	fmt	0.442	+31.1	fmt	0.394	+26.2	fmt
C_L1r_X1	0.115	fmt	-	-	-	0.226	+11.0	fmt	0.249	+13.3	fmt	0.278	+16.2	fmt	0.275	+16.0	fmt
C_X1_S1	0.125	fmt	-	-	-	0.317	+19.2	fmt	0.290	+16.5	fmt	0.339	+21.4	fmt	0.372	+24.7	fmt
C_S2_X1	0.176	fmt	-	-	-	0.372	+19.6	fmt	0.355	+17.9	fmt	0.403	+22.7	fmt	0.395	+21.9	fmt
<i>unseen-containing stacks, pooled (3)</i>	0.139	fmt	-	-	-	0.305	+16.6	fmt	0.298	+15.9	fmt	0.340	+20.1	fmt	0.348	+20.9	fmt
C_L1r_X1m	0.128	fmt	-	-	-	0.261	+13.3	fmt	0.275	+14.6	fmt	0.294	+16.5	fmt	0.317	+18.9	fmt
C_S1_X1s	0.176	fmt	-	-	-	0.390	+21.4	fmt	0.382	+20.6	fmt	0.415	+23.9	fmt	0.411	+23.5	fmt
C3_L1r_S1_X1	0.118	fmt	-	-	-	0.256	+13.8	fmt	0.262	+14.4	fmt	0.297	+17.9	fmt	0.293	+17.5	fmt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.363	100	-	-	-	0.260	-10.3	72	fmt	fmt	fmt	0.284	-7.8	78	0.269	-9.4	74
L1b	0.327	90	-	-	-	0.253	-7.4	70	fmt	fmt	fmt	0.287	-4.0	79	0.227	-10.0	63
L1r	0.339	93	-	-	-	0.255	-8.4	70	fmt	fmt	fmt	0.256	-8.3	71	0.241	-9.8	67
L2	0.357	99	-	-	-	0.280	-7.7	77	fmt	fmt	fmt	0.276	-8.1	76	0.268	-8.9	74
S1	0.323	89	-	-	-	0.305	-1.8	84	fmt	fmt	fmt	0.264	-5.8	73	0.246	-7.7	68
S2	0.350	96	-	-	-	0.354	+0.4	98	fmt	fmt	fmt	0.306	-4.4	84	0.278	-7.2	77
X1	0.303	84	-	-	-	0.253	-5.0	70	fmt	fmt	fmt	0.249	-5.3	69	0.207	-9.6	57
C_L1b_S1	0.257	71	-	-	-	0.272	+1.5	75	fmt	fmt	fmt	0.258	+0.1	71	fmt	fmt	fmt
C_L1r_S1	0.304	84	-	-	-	0.285	-1.8	79	fmt	fmt	fmt	0.200	-10.4	55	0.254	-4.9	70
C_S1_L1r	0.292	80	-	-	-	0.307	+1.5	85	fmt	fmt	fmt	0.216	-7.6	60	0.264	-2.8	73
C_L2_S4	0.344	95	-	-	-	0.291	-5.3	80	fmt	fmt	fmt	0.273	-7.1	75	0.258	-8.6	71
C_L1r_S3	0.333	92	-	-	-	0.327	-0.7	90	fmt	fmt	fmt	0.256	-7.7	71	0.269	-6.5	74
C_S4_S3	0.352	97	-	-	-	0.345	-0.7	95	fmt	fmt	fmt	0.301	-5.1	83	0.276	-7.6	76
<i>seen stacks, pooled (6)</i>	0.314	86	-	-	-	0.305	-0.9	84	fmt	fmt	fmt	0.251	-6.3	69	fmt	fmt	fmt
C3_L1r_S3_S4	0.325	90	-	-	-	0.330	+0.5	91	fmt	fmt	fmt	0.251	-7.4	69	0.272	-5.3	75
C3_S1_S3_S4	0.330	91	-	-	-	0.323	-0.7	89	fmt	fmt	fmt	0.288	-4.2	79	0.270	-6.0	74
C3_L1r_S1_S4	0.291	80	-	-	-	0.311	+2.0	86	fmt	fmt	fmt	0.199	-9.2	55	0.271	-2.0	75
C4_L1r_S1_S3_S4	0.283	78	-	-	-	0.333	+5.0	92	fmt	fmt	fmt	0.214	-6.9	59	0.270	-1.3	75
<i>depth-3/4 stacks, pooled (4)</i>	0.307	85	-	-	-	0.324	+1.7	89	fmt	fmt	fmt	0.238	-6.9	66	0.271	-3.6	75
C_L1r_X1	0.275	76	-	-	-	0.240	-3.5	66	fmt	fmt	fmt	fmt	fmt	fmt	0.195	-8.0	54
C_X1_S1	0.298	82	-	-	-	0.275	-2.3	76	fmt	fmt	fmt	fmt	fmt	fmt	0.228	-7.0	63
C_S2_X1	0.378	104	-	-	-	0.366	-1.3	101	fmt	fmt	fmt	0.288	-9.1	79	0.233	-14.5	64
<i>unseen-containing stacks, pooled (3)</i>	0.317	87	-	-	-	0.294	-2.3	81	fmt	fmt	fmt	fmt	fmt	fmt	0.219	-9.8	60
C_L1r_X1m	0.269	74	-	-	-	0.230	-3.9	63	fmt	fmt	fmt	fmt	fmt	fmt	0.189	-8.0	52
C_S1_X1s	0.346	95	-	-	-	0.302	-4.4	83	fmt	fmt	fmt	0.268	-7.8	74	0.250	-9.6	69
C3_L1r_S1_X1	0.304	84	-	-	-	0.296	-0.8	82	fmt	fmt	fmt	fmt	fmt	fmt	0.237	-6.6	65

Table 20. **Master grid, Llama-3.1-8B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.258	100	0.320	+6.2	124	0.448	+19.0	174	0.426	+16.8	165	0.445	+18.7	172	0.459	+20.1	178
L1b	0.222	86	0.276	+5.4	107	0.361	+13.9	140	0.387	+16.5	150	0.407	+18.5	158	0.369	+14.7	143
L1r	0.242	94	0.284	+4.3	110	0.387	+14.6	150	0.387	+14.5	150	0.405	+16.3	157	0.378	+13.7	147
L2	0.215	83	0.278	+6.3	108	0.391	+17.6	152	0.386	+17.1	150	0.409	+19.4	158	0.387	+17.2	150
S1	0.213	83	0.256	+4.3	99	0.395	+18.2	153	0.391	+17.7	151	0.413	+20.0	160	0.409	+19.6	158
S2	0.205	79	0.272	+6.7	106	0.413	+20.8	160	0.413	+20.8	160	0.445	+23.9	172	0.438	+23.3	170
X1	0.132	51	-	-	-	0.249	+11.7	96	0.222	+9.0	86	0.255	+12.4	99	0.332	+20.0	129
C_L1b_S1	0.152	59	-	-	-	0.279	+12.7	108	0.338	+18.7	131	0.368	+21.6	143	0.305	+15.3	118
C_L1r_S1	0.162	63	-	-	-	0.303	+14.1	118	0.352	+19.0	136	0.363	+20.1	141	0.314	+15.2	122
C_S1_L1r	0.159	62	-	-	-	0.277	+11.8	107	0.338	+17.9	131	0.354	+19.5	137	0.305	+14.6	118
C_L2_S4	0.205	79	-	-	-	0.349	+14.3	135	0.386	+18.1	149	0.401	+19.6	155	0.350	+14.5	136
C_L1r_S3	0.198	77	-	-	-	0.367	+16.9	142	0.382	+18.4	148	0.398	+19.9	154	0.369	+17.1	143
C_S4_S3	0.214	83	-	-	-	0.421	+20.7	163	0.411	+19.7	159	0.444	+23.0	172	0.431	+21.8	167
<i>seen stacks, pooled (6)</i>	0.182	70	-	-	-	0.332	+15.1	129	0.368	+18.6	142	0.388	+20.6	150	0.346	+16.4	134
C3_L1r_S3_S4	0.183	71	-	-	-	0.329	+14.6	128	0.374	+19.1	145	0.391	+20.8	152	0.363	+18.0	141
C3_S1_S3_S4	0.160	62	-	-	-	0.359	+19.8	139	0.376	+21.5	146	0.393	+23.2	152	0.378	+21.8	147
C3_L1r_S1_S4	0.160	62	-	-	-	0.279	+11.8	108	0.342	+18.2	133	0.358	+19.7	139	0.299	+13.9	116
C4_L1r_S1_S3_S4	0.140	54	-	-	-	0.263	+12.4	102	0.335	+19.6	130	0.358	+21.8	139	0.294	+15.4	114
<i>depth-3/4 stacks, pooled (4)</i>	0.161	62	-	-	-	0.308	+14.7	119	0.357	+19.6	138	0.375	+21.4	145	0.334	+17.3	129
C_L1r_X1	0.100	39	-	-	-	0.165	+6.4	64	0.181	+8.1	70	0.207	+10.6	80	0.257	+15.7	100
C_X1_S1	0.130	50	-	-	-	0.271	+14.1	105	0.225	+9.5	87	0.253	+12.3	98	0.300	+17.0	116
C_S2_X1	0.170	66	-	-	-	0.296	+12.6	115	0.284	+11.4	110	0.317	+14.7	123	0.360	+19.0	139
<i>unseen-containing stacks, pooled (3)</i>	0.134	52	-	-	-	0.244	+11.1	95	0.230	+9.6	89	0.259	+12.5	100	0.306	+17.2	118
C_L1r_X1m	0.115	45	-	-	-	0.184	+6.9	71	0.216	+10.1	84	0.232	+11.7	90	0.291	+17.6	113
C_S1_X1s	0.193	75	-	-	-	0.341	+14.8	132	0.299	+10.6	116	0.335	+14.2	130	0.355	+16.2	138
C3_L1r_S1_X1	0.105	41	-	-	-	0.194	+8.9	75	0.207	+10.2	80	0.224	+11.9	87	0.251	+14.6	97
<i>Backward: input prediction, graded by execution</i>																	
L0	0.409	100	-	-	-	0.335	-7.4	82	0.320	-9.0	78	0.332	-7.8	81	0.365	-4.4	89
L1b	0.371	91	-	-	-	0.297	-7.4	73	0.280	-9.2	68	0.285	-8.6	70	0.297	-7.5	72
L1r	0.373	91	-	-	-	0.312	-6.1	76	0.290	-8.3	71	0.297	-7.5	73	0.316	-5.7	77
L2	0.376	92	-	-	-	0.297	-7.8	73	0.284	-9.2	69	0.294	-8.1	72	0.319	-5.7	78
S1	0.375	92	-	-	-	0.325	-5.0	79	0.285	-9.0	70	0.291	-8.4	71	0.360	-1.5	88
S2	0.378	92	-	-	-	0.345	-3.4	84	0.326	-5.2	80	0.325	-5.3	79	0.366	-1.3	89
X1	0.263	64	-	-	-	0.267	+0.3	65	0.211	-5.3	51	0.228	-3.5	56	0.302	+3.9	74
C_L1b_S1	0.287	70	-	-	-	0.253	-3.4	62	0.274	-1.4	67	0.258	-2.9	63	0.297	+1.0	72
C_L1r_S1	0.297	73	-	-	-	0.281	-1.7	69	0.280	-1.8	68	0.283	-1.4	69	0.271	-2.6	66
C_S1_L1r	0.303	74	-	-	-	0.269	-3.4	66	0.274	-2.9	67	0.274	-2.9	67	0.290	-1.3	71
C_L2_S4	0.320	78	-	-	-	0.275	-4.6	67	0.269	-5.2	66	0.286	-3.4	70	0.309	-1.1	75
C_L1r_S3	0.342	84	-	-	-	0.297	-4.5	73	0.299	-4.3	73	0.303	-3.9	74	0.308	-3.4	75
C_S4_S3	0.367	90	-	-	-	0.333	-3.4	81	0.313	-5.4	76	0.329	-3.8	80	0.362	-0.5	88
<i>seen stacks, pooled (6)</i>	0.319	78	-	-	-	0.284	-3.5	70	0.285	-3.5	70	0.289	-3.1	71	0.306	-1.3	75
C3_L1r_S3_S4	0.309	75	-	-	-	0.271	-3.8	66	0.303	-0.5	74	0.300	-0.8	73	0.300	-0.9	73
C3_S1_S3_S4	0.353	86	-	-	-	0.311	-4.2	76	0.289	-6.3	71	0.281	-7.2	69	0.340	-1.3	83
C3_L1r_S1_S4	0.302	74	-	-	-	0.270	-3.2	66	0.262	-4.0	64	0.271	-3.1	66	0.270	-3.2	66
C4_L1r_S1_S3_S4	0.281	69	-	-	-	0.234	-4.7	57	0.280	-0.2	68	0.279	-0.2	68	0.259	-2.2	63
<i>depth-3/4 stacks, pooled (4)</i>	0.311	76	-	-	-	0.272	-4.0	66	0.284	-2.8	69	0.283	-2.8	69	0.292	-1.9	71
C_L1r_X1	0.218	53	-	-	-	0.220	+0.2	54	0.193	-2.6	47	0.239	+2.1	58	0.265	+4.7	65
C_X1_S1	0.291	71	-	-	-	0.283	-0.7	69	0.210	-8.1	51	0.220	-7.1	54	0.302	+1.1	74
C_S2_X1	0.323	79	-	-	-	0.288	-3.4	70	0.229	-9.3	56	0.241	-8.1	59	0.306	-1.6	75
<i>unseen-containing stacks, pooled (3)</i>	0.277	68	-	-	-	0.264	-1.3	64	0.211	-6.7	51	0.233	-4.4	57	0.291	+1.4	71
C_L1r_X1m	0.232	57	-	-	-	0.210	-2.2	51	0.217	-1.5	53	0.245	+1.3	60	0.258	+2.7	63
C_S1_X1s	0.357	87	-	-	-	0.287	-7.0	70	0.205	-15.2	50	0.212	-14.5	52	0.330	-2.7	81
C3_L1r_S1_X1	0.268	65	-	-	-	0.234	-3.4	57	0.188	-8.0	46	0.194	-7.4	47	0.243	-2.5	59

Table 21. **Master grid, StarCoder2-15B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fnt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base	acc	acc	base_1shot	%	acc	tuned_L0	%	acc	mono_all	%	acc	cons_lam3	%	acc	tuned_X1	%
<i>Forward: output prediction</i>																	
L0	fnt	fnt	fnt	fnt	fnt	0.542	fnt	fnt	0.529	fnt	fnt	0.532	fnt	fnt	0.513	fnt	fnt
L1b	fnt	fnt	fnt	fnt	fnt	0.431	fnt	fnt	0.487	fnt	fnt	0.510	fnt	fnt	0.402	fnt	fnt
L1r	fnt	fnt	fnt	fnt	fnt	0.483	fnt	fnt	0.485	fnt	fnt	0.509	fnt	fnt	0.448	fnt	fnt
L2	fnt	fnt	fnt	fnt	fnt	0.479	fnt	fnt	0.486	fnt	fnt	0.503	fnt	fnt	0.437	fnt	fnt
S1	fnt	fnt	fnt	fnt	fnt	0.508	fnt	fnt	0.487	fnt	fnt	0.501	fnt	fnt	0.470	fnt	fnt
S2	fnt	fnt	fnt	fnt	fnt	0.512	fnt	fnt	0.514	fnt	fnt	0.539	fnt	fnt	0.501	fnt	fnt
X1	fnt	fnt	-	-	-	0.347	fnt	fnt	0.320	fnt	fnt	0.343	fnt	fnt	0.381	fnt	fnt
C_L1b_S1	fnt	fnt	-	-	-	0.360	fnt	fnt	0.416	fnt	fnt	0.456	fnt	fnt	0.336	fnt	fnt
C_L1r_S1	fnt	fnt	-	-	-	0.393	fnt	fnt	0.423	fnt	fnt	0.440	fnt	fnt	0.377	fnt	fnt
C_S1_L1r	fnt	fnt	-	-	-	0.377	fnt	fnt	0.406	fnt	fnt	0.431	fnt	fnt	0.370	fnt	fnt
C_L2_S4	fnt	fnt	-	-	-	0.454	fnt	fnt	0.470	fnt	fnt	0.508	fnt	fnt	0.421	fnt	fnt
C_L1r_S3	fnt	fnt	-	-	-	0.473	fnt	fnt	0.483	fnt	fnt	0.504	fnt	fnt	0.435	fnt	fnt
C_S4_S3	fnt	fnt	-	-	-	0.511	fnt	fnt	0.516	fnt	fnt	0.539	fnt	fnt	0.490	fnt	fnt
seen stacks, pooled (6)	fnt	fnt	-	-	-	0.428	fnt	fnt	0.452	fnt	fnt	0.480	fnt	fnt	0.405	fnt	fnt
C3_L1r_S3_S4	fnt	fnt	-	-	-	0.448	fnt	fnt	0.484	fnt	fnt	0.503	fnt	fnt	0.413	fnt	fnt
C3_S1_S3_S4	fnt	fnt	-	-	-	0.476	fnt	fnt	0.465	fnt	fnt	0.493	fnt	fnt	0.459	fnt	fnt
C3_L1r_S1_S4	fnt	fnt	-	-	-	0.370	fnt	fnt	0.417	fnt	fnt	0.439	fnt	fnt	0.366	fnt	fnt
C4_L1r_S1_S3_S4	fnt	fnt	-	-	-	0.357	fnt	fnt	0.412	fnt	fnt	0.437	fnt	fnt	0.353	fnt	fnt
depth-3/4 stacks, pooled (4)	fnt	fnt	-	-	-	0.412	fnt	fnt	0.444	fnt	fnt	0.468	fnt	fnt	0.398	fnt	fnt
C_L1r_X1	fnt	fnt	-	-	-	0.248	fnt	fnt	0.274	fnt	fnt	0.286	fnt	fnt	0.288	fnt	fnt
C_X1_S1	fnt	fnt	-	-	-	0.357	fnt	fnt	0.322	fnt	fnt	0.342	fnt	fnt	0.401	fnt	fnt
C_S2_X1	fnt	fnt	-	-	-	0.403	fnt	fnt	0.375	fnt	fnt	0.397	fnt	fnt	0.420	fnt	fnt
unseen-containing stacks, pooled (3)	fnt	fnt	-	-	-	0.336	fnt	fnt	0.324	fnt	fnt	0.341	fnt	fnt	0.370	fnt	fnt
C_L1r_X1m	fnt	fnt	-	-	-	0.307	fnt	fnt	0.323	fnt	fnt	0.338	fnt	fnt	0.342	fnt	fnt
C_S1_X1s	fnt	fnt	-	-	-	0.434	fnt	fnt	0.408	fnt	fnt	0.434	fnt	fnt	0.435	fnt	fnt
C3_L1r_S1_X1	fnt	fnt	-	-	-	0.278	fnt	fnt	0.298	fnt	fnt	0.302	fnt	fnt	0.306	fnt	fnt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.366	100	-	-	-	0.411	+4.4	112	0.311	-5.6	85	0.399	+3.2	109	0.393	+2.7	107
L1b	fnt	fnt	-	-	-	0.316	fnt	86	0.301	fnt	82	0.395	fnt	108	fnt	fnt	fnt
L1r	fnt	fnt	-	-	-	0.354	fnt	97	0.273	fnt	75	0.369	fnt	101	0.298	fnt	81
L2	0.355	97	-	-	-	0.391	+3.7	107	0.297	-5.7	81	0.379	+2.4	103	0.363	+0.8	99
S1	fnt	fnt	-	-	-	0.361	fnt	98	fnt	fnt	fnt	0.364	fnt	99	0.342	fnt	93
S2	fnt	fnt	-	-	-	0.359	fnt	98	0.300	fnt	82	0.384	fnt	105	fnt	fnt	fnt
X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.281	fnt	77	fnt	fnt	fnt
C_L1b_S1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.317	fnt	87	fnt	fnt	fnt
C_L1r_S1	fnt	fnt	-	-	-	0.301	fnt	82	fnt	fnt	fnt	0.303	fnt	83	fnt	fnt	fnt
C_S1_L1r	fnt	fnt	-	-	-	0.311	fnt	85	fnt	fnt	fnt	0.303	fnt	83	fnt	fnt	fnt
C_L2_S4	fnt	fnt	-	-	-	0.342	fnt	93	fnt	fnt	fnt	0.353	fnt	96	0.290	fnt	79
C_L1r_S3	fnt	fnt	-	-	-	0.300	fnt	82	0.282	fnt	77	0.368	fnt	100	fnt	fnt	fnt
C_S4_S3	fnt	fnt	-	-	-	0.368	fnt	101	0.301	fnt	82	0.386	fnt	105	fnt	fnt	fnt
seen stacks, pooled (6)	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.338	fnt	92	fnt	fnt	fnt
C3_L1r_S3_S4	fnt	fnt	-	-	-	0.294	fnt	80	fnt	fnt	fnt	0.339	fnt	92	fnt	fnt	fnt
C3_S1_S3_S4	fnt	fnt	-	-	-	0.337	fnt	92	fnt	fnt	fnt	0.333	fnt	91	fnt	fnt	fnt
C3_L1r_S1_S4	fnt	fnt	-	-	-	0.293	fnt	80	fnt	fnt	fnt	0.270	fnt	74	fnt	fnt	fnt
C4_L1r_S1_S3_S4	fnt	fnt	-	-	-	0.289	fnt	79	fnt	fnt	fnt	0.273	fnt	74	fnt	fnt	fnt
depth-3/4 stacks, pooled (4)	fnt	fnt	-	-	-	0.304	fnt	83	fnt	fnt	fnt	0.304	fnt	83	fnt	fnt	fnt
C_L1r_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_X1_S1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_S2_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
unseen-containing stacks, pooled (3)	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C_L1r_X1m	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	0.234	fnt	64	fnt	fnt	fnt
C_S1_X1s	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt
C3_L1r_S1_X1	fnt	fnt	-	-	-	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt	fnt

Table 22. Master grid, Gemma-3-12B. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	fmt	fmt	fmt	fmt	fmt	0.520	fmt	fmt	0.507	fmt	fmt	0.502	fmt	fmt	0.506	fmt	fmt
L1b	fmt	fmt	fmt	fmt	fmt	0.441	fmt	fmt	0.475	fmt	fmt	0.488	fmt	fmt	0.402	fmt	fmt
L1r	fmt	fmt	fmt	fmt	fmt	0.459	fmt	fmt	0.463	fmt	fmt	0.472	fmt	fmt	0.447	fmt	fmt
L2	fmt	fmt	fmt	fmt	fmt	0.456	fmt	fmt	0.471	fmt	fmt	0.478	fmt	fmt	0.441	fmt	fmt
S1	fmt	fmt	fmt	fmt	fmt	0.456	fmt	fmt	0.457	fmt	fmt	0.483	fmt	fmt	0.452	fmt	fmt
S2	fmt	fmt	fmt	fmt	fmt	0.500	fmt	fmt	0.503	fmt	fmt	0.504	fmt	fmt	0.489	fmt	fmt
X1	0.152	fmt	-	-	-	0.310	+15.7	fmt	0.263	+11.0	fmt	0.289	+13.7	fmt	0.376	+22.3	fmt
C_L1b_S1	fmt	fmt	-	-	-	0.333	fmt	fmt	0.406	fmt	fmt	0.420	fmt	fmt	0.330	fmt	fmt
C_L1r_S1	fmt	fmt	-	-	-	0.336	fmt	fmt	0.389	fmt	fmt	0.416	fmt	fmt	0.344	fmt	fmt
C_S1_L1r	fmt	fmt	-	-	-	0.334	fmt	fmt	0.393	fmt	fmt	0.414	fmt	fmt	0.329	fmt	fmt
C_L2_S4	fmt	fmt	-	-	-	0.438	fmt	fmt	0.460	fmt	fmt	0.469	fmt	fmt	0.424	fmt	fmt
C_L1r_S3	fmt	fmt	-	-	-	0.438	fmt	fmt	0.462	fmt	fmt	0.469	fmt	fmt	0.423	fmt	fmt
C_S4_S3	fmt	fmt	-	-	-	0.491	fmt	fmt	0.506	fmt	fmt	0.503	fmt	fmt	0.481	fmt	fmt
<i>seen stacks, pooled (6)</i>	fmt	fmt	-	-	-	0.395	fmt	fmt	0.436	fmt	fmt	0.448	fmt	fmt	0.388	fmt	fmt
C3_L1r_S3_S4	fmt	fmt	-	-	-	0.419	fmt	fmt	0.454	fmt	fmt	0.467	fmt	fmt	0.417	fmt	fmt
C3_S1_S3_S4	fmt	fmt	-	-	-	0.425	fmt	fmt	0.452	fmt	fmt	0.465	fmt	fmt	0.431	fmt	fmt
C3_L1r_S1_S4	fmt	fmt	-	-	-	0.320	fmt	fmt	0.390	fmt	fmt	0.415	fmt	fmt	0.330	fmt	fmt
C4_L1r_S1_S3_S4	fmt	fmt	-	-	-	0.302	fmt	fmt	0.394	fmt	fmt	0.413	fmt	fmt	0.323	fmt	fmt
<i>depth-3/4 stacks, pooled (4)</i>	fmt	fmt	-	-	-	0.366	fmt	fmt	0.423	fmt	fmt	0.440	fmt	fmt	0.375	fmt	fmt
C_L1r_X1	0.102	fmt	-	-	-	0.195	+9.3	fmt	0.206	+10.4	fmt	0.224	+12.2	fmt	0.285	+18.3	fmt
C_X1_S1	0.172	fmt	-	-	-	0.317	+14.6	fmt	0.271	+10.0	fmt	0.296	+12.5	fmt	0.358	+18.6	fmt
C_S2_X1	0.201	fmt	-	-	-	0.368	+16.6	fmt	0.349	+14.8	fmt	0.364	+16.3	fmt	0.412	+21.1	fmt
<i>unseen-containing stacks, pooled (3)</i>	0.158	fmt	-	-	-	0.293	+13.5	fmt	0.275	+11.7	fmt	0.295	+13.7	fmt	0.352	+19.3	fmt
C_L1r_X1m	fmt	fmt	-	-	-	0.220	fmt	fmt	0.235	fmt	fmt	0.244	fmt	fmt	0.312	fmt	fmt
C_S1_X1s	0.255	fmt	-	-	-	0.395	+14.0	fmt	0.374	+12.0	fmt	0.395	+14.0	fmt	0.401	+14.7	fmt
C3_L1r_S1_X1	0.136	fmt	-	-	-	0.225	+8.9	fmt	0.236	+10.0	fmt	0.258	+12.2	fmt	0.260	+12.4	fmt
<i>Backward: input prediction, graded by execution</i>																	
L0	0.453	100	-	-	-	0.440	-1.3	97	0.447	-0.6	99	0.355	-9.8	78	0.469	+1.6	104
L1b	0.405	89	-	-	-	0.353	-5.2	78	0.427	+2.2	94	fmt	fmt	fmt	0.395	-1.0	87
L1r	0.420	93	-	-	-	0.394	-2.6	87	0.411	-0.9	91	fmt	fmt	fmt	0.428	+0.8	94
L2	0.403	89	-	-	-	0.378	-2.6	83	0.405	+0.2	89	fmt	fmt	fmt	0.431	+2.8	95
S1	0.424	94	-	-	-	0.375	-4.9	83	0.421	-0.2	93	fmt	fmt	fmt	0.461	+3.7	102
S2	0.443	98	-	-	-	0.444	+0.1	98	0.457	+1.4	101	fmt	fmt	fmt	0.454	+1.1	100
X1	0.333	74	-	-	-	0.305	-2.9	67	0.333	+0.0	74	fmt	fmt	fmt	0.393	+6.0	87
C_L1b_S1	0.349	77	-	-	-	0.317	-3.3	70	0.366	+1.7	81	fmt	fmt	fmt	0.370	+2.1	82
C_L1r_S1	0.356	79	-	-	-	0.319	-3.7	70	0.379	+2.2	84	fmt	fmt	fmt	0.397	+4.1	88
C_S1_L1r	0.337	74	-	-	-	0.319	-1.8	70	0.379	+4.1	84	fmt	fmt	fmt	0.407	+6.9	90
C_L2_S4	0.387	85	-	-	-	0.381	-0.6	84	0.416	+2.9	92	fmt	fmt	fmt	0.427	+4.0	94
C_L1r_S3	0.427	94	-	-	-	0.402	-2.6	89	0.423	-0.5	93	fmt	fmt	fmt	0.438	+1.1	97
C_S4_S3	0.434	96	-	-	-	0.438	+0.4	97	0.462	+2.8	102	fmt	fmt	fmt	0.469	+3.5	104
<i>seen stacks, pooled (6)</i>	0.382	84	-	-	-	0.363	-1.9	80	0.404	+2.2	89	fmt	fmt	fmt	0.418	+3.6	92
C3_L1r_S3_S4	0.398	88	-	-	-	0.386	-1.2	85	0.420	+2.2	93	fmt	fmt	fmt	0.447	+4.9	99
C3_S1_S3_S4	0.387	86	-	-	-	0.354	-3.3	78	0.419	+3.1	92	fmt	fmt	fmt	0.440	+5.2	97
C3_L1r_S1_S4	0.347	77	-	-	-	0.302	-4.5	67	0.376	+3.0	83	fmt	fmt	fmt	0.377	+3.0	83
C4_L1r_S1_S3_S4	0.329	73	-	-	-	0.293	-3.7	65	0.364	+3.5	80	fmt	fmt	fmt	0.399	+6.9	88
<i>depth-3/4 stacks, pooled (4)</i>	0.365	81	-	-	-	0.334	-3.2	74	0.395	+2.9	87	fmt	fmt	fmt	0.416	+5.0	92
C_L1r_X1	0.258	57	-	-	-	0.242	-1.6	53	0.279	+2.1	62	fmt	fmt	fmt	0.329	+7.2	73
C_X1_S1	0.348	77	-	-	-	0.282	-6.5	62	0.331	-1.7	73	fmt	fmt	fmt	0.410	+6.2	91
C_S2_X1	0.360	79	-	-	-	0.339	-2.0	75	0.357	-0.3	79	fmt	fmt	fmt	0.398	+3.8	88
<i>unseen-containing stacks, pooled (3)</i>	0.322	71	-	-	-	0.288	-3.4	64	0.322	+0.1	71	fmt	fmt	fmt	0.379	+5.7	84
C_L1r_X1m	0.273	60	-	-	-	0.269	-0.4	59	0.323	+5.0	71	fmt	fmt	fmt	0.349	+7.6	77
C_S1_X1s	0.412	91	-	-	-	0.371	-4.1	82	0.383	-3.0	84	fmt	fmt	fmt	0.448	+3.6	99
C3_L1r_S1_X1	0.279	62	-	-	-	0.226	-5.3	50	0.270	-0.9	60	fmt	fmt	fmt	0.337	+5.8	74

Table 23. **Master grid, CodeGemma-7B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.237	100	fmt	fmt	fmt	0.465	+22.9	197	0.439	+20.2	186	0.445	+20.8	188	0.462	+22.6	195
L1b	0.185	78	fmt	fmt	fmt	0.382	+19.7	161	0.425	+24.0	180	0.435	+25.1	184	0.345	+16.0	146
L1r	0.213	90	fmt	fmt	fmt	0.399	+18.6	169	0.417	+20.4	176	0.417	+20.4	176	0.390	+17.7	165
L2	0.208	88	fmt	fmt	fmt	0.412	+20.4	174	0.422	+21.4	178	0.422	+21.4	178	0.392	+18.3	166
S1	0.195	82	0.212	+1.7	90	0.405	+21.0	171	0.415	+22.1	176	0.421	+22.6	178	0.423	+22.8	179
S2	0.198	84	fmt	fmt	fmt	0.450	+25.2	190	0.429	+23.1	181	0.448	+25.0	189	0.444	+24.6	188
X1	0.143	61	-	-	-	0.271	+12.8	115	0.251	+10.8	106	0.247	+10.4	104	0.348	+20.4	147
C_L1b_S1	0.148	63	-	-	-	0.286	+13.7	121	0.372	+22.3	157	0.368	+21.9	156	0.287	+13.9	121
C_L1r_S1	0.144	61	-	-	-	0.315	+17.2	133	0.373	+22.9	158	0.367	+22.3	155	0.304	+16.0	129
C_S1_L1r	0.132	56	-	-	-	0.288	+15.6	122	0.363	+23.1	154	0.361	+22.8	153	0.299	+16.7	127
C_L2_S4	0.176	75	-	-	-	0.391	+21.5	165	0.423	+24.7	179	0.421	+24.5	178	0.374	+19.8	158
C_L1r_S3	0.183	77	-	-	-	0.396	+21.3	168	0.415	+23.2	175	0.410	+22.7	173	0.375	+19.2	159
C_S4_S3	0.202	85	-	-	-	0.448	+24.7	189	0.430	+22.9	182	0.441	+23.9	186	0.446	+24.5	189
seen stacks, pooled (6)	0.164	69	-	-	-	0.354	+19.0	150	0.396	+23.2	167	0.395	+23.0	167	0.348	+18.4	147
C3_L1r_S3_S4	0.157	66	-	-	-	0.376	+21.8	159	0.407	+25.0	172	0.411	+25.4	174	0.361	+20.3	152
C3_S1_S3_S4	0.158	67	-	-	-	0.367	+20.9	155	0.395	+23.7	167	0.405	+24.7	171	0.392	+23.4	166
C3_L1r_S1_S4	0.126	53	-	-	-	0.289	+16.3	122	0.367	+24.1	155	0.366	+23.9	155	0.301	+17.5	127
C4_L1r_S1_S3_S4	0.124	52	-	-	-	0.271	+14.7	114	0.361	+23.7	153	0.359	+23.5	152	0.285	+16.1	120
depth-3/4 stacks, pooled (4)	0.141	60	-	-	-	0.326	+18.4	138	0.383	+24.1	162	0.385	+24.4	163	0.335	+19.3	141
C_L1r_X1	0.118	50	-	-	-	0.194	+7.6	82	0.212	+9.4	90	0.197	+7.9	83	0.245	+12.8	104
C_X1_S1	0.144	61	-	-	-	0.285	+14.0	120	0.261	+11.6	110	0.247	+10.3	105	0.339	+19.4	143
C_S2_X1	0.188	80	-	-	-	0.344	+15.6	146	0.323	+13.5	136	0.326	+13.8	138	0.379	+19.1	160
unseen-containing stacks, pooled (3)	0.150	63	-	-	-	0.274	+12.4	116	0.265	+11.5	112	0.257	+10.7	109	0.321	+17.1	136
C_L1r_X1m	0.130	55	-	-	-	0.214	+8.4	90	0.256	+12.5	108	0.234	+10.4	99	0.288	+15.8	122
C_S1_X1s	0.186	79	-	-	-	0.338	+15.2	143	0.351	+16.5	148	0.339	+15.3	143	0.386	+20.0	163
C3_L1r_S1_X1	0.134	57	-	-	-	0.218	+8.4	92	0.241	+10.6	102	0.246	+11.2	104	0.254	+11.9	107
<i>Backward: input prediction, graded by execution</i>																	
L0	0.266	100	-	-	-	0.364	+9.8	137	0.324	+5.8	122	fmt	fmt	fmt	0.256	-1.0	96
L1b	0.268	101	-	-	-	0.309	+4.1	116	0.298	+3.1	112	fmt	fmt	fmt	0.193	-7.4	73
L1r	0.251	94	-	-	-	0.333	+8.3	125	0.301	+5.0	113	fmt	fmt	fmt	0.225	-2.6	85
L2	0.239	90	-	-	-	0.327	+8.7	123	0.291	+5.1	109	fmt	fmt	fmt	0.220	-1.9	83
S1	0.234	88	-	-	-	0.357	+12.3	134	0.337	+10.3	127	fmt	fmt	fmt	0.231	-0.3	87
S2	0.270	102	-	-	-	0.372	+10.2	140	0.351	+8.0	132	fmt	fmt	fmt	0.248	-2.3	93
X1	0.190	72	-	-	-	0.260	+7.0	98	0.217	+2.7	82	fmt	fmt	fmt	0.235	+4.4	88
C_L1b_S1	0.229	86	-	-	-	0.274	+4.5	103	0.276	+4.7	104	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_S1	0.170	64	-	-	-	0.289	+11.9	109	0.290	+12.0	109	fmt	fmt	fmt	0.208	+3.8	78
C_S1_L1r	0.168	63	-	-	-	0.284	+11.6	107	0.316	+14.8	119	fmt	fmt	fmt	0.203	+3.4	76
C_L2_S4	0.239	90	-	-	-	0.318	+8.0	120	0.299	+6.0	112	fmt	fmt	fmt	0.238	-0.1	90
C_L1r_S3	0.223	84	-	-	-	0.326	+10.4	123	0.314	+9.1	118	fmt	fmt	fmt	0.223	-0.0	84
C_S4_S3	0.257	97	-	-	-	0.372	+11.5	140	0.357	+10.0	134	fmt	fmt	fmt	0.242	-1.5	91
seen stacks, pooled (6)	0.214	81	-	-	-	0.310	+9.6	117	0.308	+9.4	116	fmt	fmt	fmt	fmt	fmt	fmt
C3_L1r_S3_S4	0.187	70	-	-	-	0.318	+13.1	120	0.312	+12.5	118	fmt	fmt	fmt	0.221	+3.4	83
C3_S1_S3_S4	0.231	87	-	-	-	0.338	+10.7	127	0.332	+10.0	125	fmt	fmt	fmt	0.234	+0.3	88
C3_L1r_S1_S4	0.173	65	-	-	-	0.274	+10.0	103	0.286	+11.3	108	fmt	fmt	fmt	0.191	+1.8	72
C4_L1r_S1_S3_S4	0.150	56	-	-	-	0.267	+11.7	101	0.290	+14.0	109	fmt	fmt	fmt	0.177	+2.7	67
depth-3/4 stacks, pooled (4)	0.185	70	-	-	-	0.299	+11.4	113	0.305	+12.0	115	fmt	fmt	fmt	0.206	+2.0	77
C_L1r_X1	fmt	fmt	-	-	-	0.208	fmt	78	0.174	fmt	66	fmt	fmt	fmt	0.178	fmt	67
C_X1_S1	0.202	76	-	-	-	0.303	+10.1	114	0.209	+0.6	79	fmt	fmt	fmt	0.227	+2.5	86
C_S2_X1	0.227	85	-	-	-	0.300	+7.3	113	0.241	+1.4	91	fmt	fmt	fmt	fmt	fmt	fmt
unseen-containing stacks, pooled (3)	fmt	fmt	-	-	-	0.271	fmt	102	0.208	fmt	78	fmt	fmt	fmt	fmt	fmt	fmt
C_L1r_X1m	fmt	fmt	-	-	-	0.231	fmt	87	0.211	fmt	79	fmt	fmt	fmt	0.205	fmt	77
C_S1_X1s	0.230	86	-	-	-	0.322	+9.3	121	0.230	+0.1	87	fmt	fmt	fmt	0.243	+1.4	92
C3_L1r_S1_X1	0.168	63	-	-	-	0.230	+6.2	87	fmt	fmt	fmt	fmt	fmt	fmt	0.191	+2.3	72

Table 24. **Master grid, Granite-3.1-8B**. Every condition and every stack, forward (output prediction) and backward (input prediction). Per system: *acc* raw accuracy; Δ points against the untuned model on the same condition; % accuracy as a percentage of the untuned model’s clean-code accuracy in the same direction, i.e. how much of the original accuracy the system brings back (100 = the untuned model on clean code). *fmt*: format-failure rate above 0.25, the cell measures the prompt contract rather than the task and enters no mean. Pooled rows average the stacks above them. ‘-’: cell not run.

Condition	untuned		ICL			clean LoRA			breadth			anchored			family		
	base		base_1shot			tuned_L0			mono_all			cons_lam3			tuned_X1		
	acc	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%	acc	Δ	%
<i>Forward: output prediction</i>																	
L0	0.285	100	0.293	+0.8	103	0.430	+14.5	151	0.416	+13.1	146	0.396	+11.1	139	0.413	+12.8	145
L1b	0.267	94	0.267	+0.0	94	0.338	+7.1	118	0.391	+12.5	137	0.381	+11.4	134	0.321	+5.4	113
L1r	0.268	94	0.260	-0.8	91	0.375	+10.8	132	0.383	+11.5	134	0.378	+11.1	133	0.376	+10.8	132
L2	0.251	88	0.248	-0.4	87	0.372	+12.0	130	0.387	+13.5	136	0.377	+12.6	132	0.366	+11.4	128
S1	0.254	89	0.253	-0.2	89	0.346	+9.1	121	0.379	+12.5	133	0.358	+10.4	126	0.350	+9.5	123
S2	0.272	96	0.276	+0.4	97	0.386	+11.4	136	0.415	+14.2	145	0.394	+12.1	138	0.398	+12.5	140
X1	0.147	51	-	-	-	0.216	+6.9	76	0.207	+6.0	73	0.188	+4.1	66	0.282	+13.5	99
C_L1b_S1	0.219	77	-	-	-	0.263	+4.4	92	0.342	+12.4	120	0.337	+11.8	118	0.261	+4.2	92
C_L1r_S1	0.204	72	-	-	-	0.279	+7.5	98	0.344	+14.0	121	0.338	+13.3	118	0.291	+8.7	102
C_S1_L1r	0.224	79	-	-	-	0.279	+5.4	98	0.326	+10.2	115	0.319	+9.5	112	0.291	+6.6	102
C_L2_S4	0.242	85	-	-	-	0.334	+9.2	117	0.376	+13.4	132	0.376	+13.4	132	0.341	+9.9	120
C_L1r_S3	0.259	91	-	-	-	0.350	+9.1	123	0.382	+12.3	134	0.376	+11.7	132	0.359	+10.1	126
C_S4_S3	0.270	95	-	-	-	0.380	+11.0	133	0.412	+14.2	144	0.389	+11.9	137	0.383	+11.3	134
<i>seen stacks, pooled (6)</i>	0.236	83	-	-	-	0.314	+7.8	110	0.364	+12.7	128	0.356	+12.0	125	0.321	+8.5	113
C3_L1r_S3_S4	0.257	90	-	-	-	0.333	+7.6	117	0.380	+12.3	133	0.375	+11.8	132	0.340	+8.3	119
C3_S1_S3_S4	0.242	85	-	-	-	0.316	+7.4	111	0.372	+13.1	131	0.358	+11.7	126	0.330	+8.9	116
C3_L1r_S1_S4	0.221	78	-	-	-	0.279	+5.7	98	0.333	+11.2	117	0.330	+10.9	116	0.283	+6.1	99
C4_L1r_S1_S3_S4	0.203	71	-	-	-	0.258	+5.5	90	0.325	+12.2	114	0.334	+13.1	117	0.270	+6.7	95
<i>depth-3/4 stacks, pooled (4)</i>	0.231	81	-	-	-	0.296	+6.6	104	0.352	+12.2	124	0.349	+11.9	123	0.306	+7.5	107
C_L1r_X1	0.136	48	-	-	-	0.157	+2.1	55	0.172	+3.6	60	0.163	+2.7	57	0.245	+11.0	86
C_X1_S1	0.153	54	-	-	-	0.222	+7.0	78	0.203	+5.0	71	0.186	+3.3	65	0.272	+12.0	96
C_S2_X1	0.209	73	-	-	-	0.263	+5.5	92	0.252	+4.4	89	0.240	+3.1	84	0.323	+11.4	113
<i>unseen-containing stacks, pooled (3)</i>	0.166	58	-	-	-	0.214	+4.8	75	0.209	+4.3	73	0.196	+3.1	69	0.280	+11.4	98
C_L1r_X1m	0.158	55	-	-	-	0.188	+3.0	66	0.204	+4.7	72	0.183	+2.6	64	0.279	+12.1	98
C_S1_X1s	0.206	72	-	-	-	0.298	+9.2	104	0.267	+6.1	94	0.269	+6.3	94	0.318	+11.2	111
C3_L1r_S1_X1	0.167	59	-	-	-	0.186	+1.8	65	0.195	+2.8	68	0.198	+3.1	70	0.238	+7.0	83
<i>Backward: input prediction, graded by execution</i>																	
L0	0.291	100	-	-	-	fmt	fmt	fmt	0.261	-3.0	90	fmt	fmt	fmt	0.272	-1.9	94
L1b	0.255	88	-	-	-	fmt	fmt	fmt	0.228	-2.7	79	fmt	fmt	fmt	0.211	-4.4	73
L1r	0.284	98	-	-	-	fmt	fmt	fmt	0.268	-1.7	92	0.150	-13.4	52	0.262	-2.3	90
L2	0.247	85	-	-	-	fmt	fmt	fmt	0.236	-1.1	81	fmt	fmt	fmt	0.222	-2.5	76
S1	0.305	105	-	-	-	fmt	fmt	fmt	0.220	-8.6	75	fmt	fmt	fmt	0.236	-6.9	81
S2	0.329	113	-	-	-	fmt	fmt	fmt	0.244	-8.5	84	fmt	fmt	fmt	0.266	-6.2	92
X1	fmt	fmt	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.207	fmt	71
C_L1b_S1	0.216	74	-	-	-	fmt	fmt	fmt	0.199	-1.8	68	fmt	fmt	fmt	0.194	-2.2	67
C_L1r_S1	0.284	98	-	-	-	0.206	-7.8	71	0.225	-5.9	77	fmt	fmt	fmt	0.234	-5.0	80
C_S1_L1r	0.262	90	-	-	-	0.208	-5.4	72	0.241	-2.2	83	fmt	fmt	fmt	0.242	-2.0	83
C_L2_S4	0.278	96	-	-	-	fmt	fmt	fmt	0.225	-5.3	78	fmt	fmt	fmt	0.213	-6.5	73
C_L1r_S3	0.296	102	-	-	-	fmt	fmt	fmt	0.240	-5.6	83	fmt	fmt	fmt	0.251	-4.5	86
C_S4_S3	0.334	115	-	-	-	fmt	fmt	fmt	0.254	-8.0	87	fmt	fmt	fmt	0.254	-8.0	87
<i>seen stacks, pooled (6)</i>	0.278	96	-	-	-	fmt	fmt	fmt	0.231	-4.8	79	fmt	fmt	fmt	0.231	-4.7	80
C3_L1r_S3_S4	0.331	114	-	-	-	fmt	fmt	fmt	0.238	-9.3	82	fmt	fmt	fmt	0.235	-9.6	81
C3_S1_S3_S4	0.323	111	-	-	-	fmt	fmt	fmt	0.214	-10.9	74	fmt	fmt	fmt	0.242	-8.1	83
C3_L1r_S1_S4	0.293	101	-	-	-	0.192	-10.0	66	0.223	-6.9	77	fmt	fmt	fmt	0.238	-5.5	82
C4_L1r_S1_S3_S4	0.305	105	-	-	-	fmt	fmt	fmt	0.216	-8.9	74	fmt	fmt	fmt	0.239	-6.5	82
<i>depth-3/4 stacks, pooled (4)</i>	0.313	108	-	-	-	fmt	fmt	fmt	0.223	-9.0	77	fmt	fmt	fmt	0.239	-7.4	82
C_L1r_X1	fmt	fmt	-	-	-	fmt	fmt	fmt	0.163	fmt	56	fmt	fmt	fmt	0.190	fmt	65
C_X1_S1	0.234	80	-	-	-	fmt	fmt	fmt	0.161	-7.3	55	fmt	fmt	fmt	0.242	+0.8	83
C_S2_X1	0.300	103	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.235	-6.5	81
<i>unseen-containing stacks, pooled (3)</i>	fmt	fmt	-	-	-	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	fmt	0.222	fmt	76
C_L1r_X1m	fmt	fmt	-	-	-	fmt	fmt	fmt	0.180	fmt	62	fmt	fmt	fmt	0.188	fmt	65
C_S1_X1s	0.282	97	-	-	-	fmt	fmt	fmt	0.173	-10.9	59	fmt	fmt	fmt	0.234	-4.8	81
C3_L1r_S1_X1	0.259	89	-	-	-	fmt	fmt	fmt	0.156	-10.3	54	fmt	fmt	fmt	0.226	-3.3	78

Table 25. **Routing, CodeLlama-7B.** MoLE mixtures of the eight per-transform experts on every condition and stack that has them: *router* = trained gate, *hard router* = its argmax, *uniform* = fixed uniform gate, *random* = gate frozen at random initialisation. Per system: raw accuracy, Δ points against the untuned model on the same items, and % of the untuned model’s clean-code accuracy. Every cell ran through the mixture engine.

Condition	untuned			router			hard router			uniform			random		
	base	acc	%	mole_router	acc	Δ	mole_hardrouter	acc	Δ	mole_uniform	acc	Δ	mole_random	acc	Δ
L0	0.258	100		0.429	+17.1	167	0.428	+17.0	167	0.428	+16.9	166	0.433	+17.5	169
L1b	0.196	76		0.381	+18.5	148	0.384	+18.8	149	0.371	+17.5	144	0.375	+17.9	146
L1r	0.205	80		0.383	+17.7	149	0.381	+17.6	148	0.390	+18.4	152	0.391	+18.6	152
L2	0.198	77		0.386	+18.8	150	0.384	+18.6	149	0.392	+19.3	152	0.393	+19.5	153
S1	0.167	65		0.395	+22.8	154	0.393	+22.6	153	0.395	+22.8	154	0.394	+22.7	153
S2	0.191	74		0.421	+23.0	164	0.421	+22.9	164	0.415	+22.3	161	0.411	+22.0	160
X1	0.118	46		0.279	+16.1	109	0.283	+16.5	110	0.283	+16.6	110	0.283	+16.5	110
C_L1b_S1	0.141	55		0.325	+18.4	126	0.324	+18.3	126	0.310	+16.8	121	0.312	+17.1	121
C_L1r_S1	0.131	51		0.346	+21.5	135	0.339	+20.8	132	0.318	+18.7	124	0.320	+18.9	125
C_S1_L1r	0.135	53		0.318	+18.4	124	0.324	+18.9	126	0.302	+16.8	118	0.303	+16.8	118
C_L2_S4	0.176	68		0.383	+20.8	149	0.379	+20.3	148	0.380	+20.4	148	0.379	+20.3	147
C_L1r_S3	0.172	67		0.391	+21.9	152	0.389	+21.7	151	0.383	+21.1	149	0.380	+20.8	148
C_S4_S3	0.185	72		0.415	+22.9	161	0.413	+22.7	161	0.409	+22.4	159	0.409	+22.4	159
C3_L1r_S3_S4	0.150	58		0.375	+22.5	146	0.376	+22.6	146	0.365	+21.5	142	0.363	+21.3	141
C3_S1_S3_S4	0.128	50		0.377	+24.8	147	0.378	+24.9	147	0.367	+23.8	143	0.367	+23.9	143
C3_L1r_S1_S4	0.126	49		0.325	+19.9	126	0.326	+20.0	127	0.306	+18.0	119	0.305	+17.9	119
C4_L1r_S1_S3_S4	0.089	35		0.334	+24.5	130	0.336	+24.7	131	0.296	+20.7	115	0.298	+20.9	116
C_L1r_X1	0.102	40		0.189	+8.7	74	0.186	+8.4	72	0.194	+9.2	76	0.190	+8.8	74
C_X1_S1	0.104	40		0.291	+18.7	113	0.286	+18.2	111	0.293	+18.9	114	0.295	+19.1	115
C_S2_X1	0.145	56		0.326	+18.1	127	0.328	+18.3	128	0.329	+18.4	128	0.329	+18.4	128
C_L1r_X1m	0.114	44		0.227	+11.3	88	0.224	+11.0	87	0.221	+10.7	86	0.221	+10.6	86
C_S1_X1s	0.148	58		0.350	+20.2	136	0.348	+20.0	135	0.344	+19.6	134	0.343	+19.5	134
C3_L1r_S1_X1	0.102	40		0.242	+14.0	94	0.245	+14.3	95	0.234	+13.2	91	0.239	+13.8	93

Table 26. **Merging, CodeLlama-7B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned			TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	base	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.257	100		0.366	+11.0	143	0.426	+16.9	166	fmt fmt	fmt		0.394	+13.7	153	0.413	+15.6	161
L1b	0.197	77		0.299	+10.1	116	0.373	+17.6	145	fmt fmt	fmt		0.331	+13.4	129	0.349	+15.1	136
L1r	0.207	81		0.323	+11.6	126	0.381	+17.4	148	fmt fmt	fmt		0.338	+13.1	132	0.352	+14.5	137
L2	0.202	79		0.317	+11.5	123	0.375	+17.4	146	fmt fmt	fmt		0.337	+13.5	131	0.355	+15.3	138
S1	0.168	66		0.322	+15.4	125	0.376	+20.8	146	fmt fmt	fmt		0.342	+17.3	133	0.365	+19.6	142
S2	0.193	75		0.340	+14.7	132	0.392	+19.9	153	fmt fmt	fmt		0.365	+17.2	142	0.379	+18.5	147
X1	0.119	46		0.234	+11.4	91	0.276	+15.7	107	fmt fmt	fmt		-	-	-	-	-	-
C_L1b_S1	0.141	55		0.245	+10.4	95	0.302	+16.0	117	fmt fmt	fmt		0.269	+12.8	105	0.280	+13.9	109
C_L1r_S1	0.131	51		0.255	+12.4	99	0.301	+17.0	117	fmt fmt	fmt		0.277	+14.6	108	0.283	+15.2	110
C_S1_L1r	0.135	53		0.245	+11.0	95	0.301	+16.6	117	fmt fmt	fmt		0.273	+13.8	106	0.282	+14.7	110
C_L2_S4	0.176	68		0.299	+12.3	116	0.363	+18.7	141	fmt fmt	fmt		0.325	+14.9	126	0.349	+17.3	136
C_L1r_S3	0.172	67		0.307	+13.5	120	0.368	+19.6	143	fmt fmt	fmt		0.326	+15.4	127	0.338	+16.6	131
C_S4_S3	0.186	72		0.341	+15.5	133	0.398	+21.2	155	fmt fmt	fmt		0.367	+18.1	143	0.385	+19.9	150
C3_L1r_S3_S4	0.150	58		0.298	+14.8	116	0.349	+19.9	136	fmt fmt	fmt		0.313	+16.3	122	0.328	+17.8	128
C3_S1_S3_S4	0.128	50		0.271	+14.3	106	0.340	+21.1	132	fmt fmt	fmt		0.295	+16.6	115	0.315	+18.7	123
C3_L1r_S1_S4	0.126	49		0.240	+11.4	94	0.296	+17.0	115	fmt fmt	fmt		0.272	+14.6	106	0.283	+15.7	110
C4_L1r_S1_S3_S4	0.089	35		0.217	+12.8	85	0.290	+20.0	113	fmt fmt	fmt		0.243	+15.4	95	0.256	+16.7	100
C_L1r_X1	0.102	40		0.171	+6.9	67	0.193	+9.1	75	fmt fmt	fmt		-	-	-	-	-	-
C_X1_S1	0.104	40		0.223	+12.0	87	0.290	+18.6	113	fmt fmt	fmt		-	-	-	-	-	-
C_S2_X1	0.145	57		0.265	+12.0	103	0.317	+17.1	123	fmt fmt	fmt		-	-	-	-	-	-
C_L1r_X1m	0.112	44		0.190	+7.8	74	0.215	+10.3	84	fmt fmt	fmt		-	-	-	-	-	-
C_S1_X1s	0.151	59		0.301	+15.0	117	0.338	+18.8	132	fmt fmt	fmt		-	-	-	-	-	-
C3_L1r_S1_X1	0.104	40		0.191	+8.7	74	0.232	+12.8	90	fmt fmt	fmt		-	-	-	-	-	-

Table 27. **Merging, Granite-3.1-8B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.285	100	0.383	+9.8	134	0.433	+14.8	152	fmt	fmt	fmt	-	-	-	-	-	-
L1b	0.266	93	0.315	+4.9	110	0.350	+8.4	123	fmt	fmt	fmt	-	-	-	-	-	-
L1r	0.268	94	0.353	+8.5	124	0.384	+11.6	135	fmt	fmt	fmt	-	-	-	-	-	-
L2	0.251	88	0.342	+9.0	120	0.378	+12.6	133	fmt	fmt	fmt	-	-	-	-	-	-
S1	0.254	89	0.323	+6.9	113	0.371	+11.7	130	fmt	fmt	fmt	-	-	-	-	-	-
S2	0.272	96	0.358	+8.5	125	0.402	+13.0	141	fmt	fmt	fmt	-	-	-	-	-	-
X1	0.147	51	0.199	+5.3	70	0.231	+8.4	81	fmt	fmt	fmt	-	-	-	-	-	-
C_L1b_S1	0.219	77	0.247	+2.9	87	0.279	+6.0	98	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_S1	0.204	72	0.266	+6.1	93	0.301	+9.7	106	fmt	fmt	fmt	-	-	-	-	-	-
C_S1_L1r	0.224	79	0.262	+3.8	92	0.293	+6.9	103	fmt	fmt	fmt	-	-	-	-	-	-
C_L2_S4	0.242	85	0.319	+7.7	112	0.351	+10.9	123	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_S3	0.259	91	0.324	+6.5	114	0.365	+10.6	128	fmt	fmt	fmt	-	-	-	-	-	-
C_S4_S3	0.270	95	0.353	+8.3	124	0.394	+12.4	138	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S3_S4	0.257	90	0.307	+5.0	108	0.348	+9.1	122	fmt	fmt	fmt	-	-	-	-	-	-
C3_S1_S3_S4	0.242	85	0.315	+7.3	111	0.342	+10.1	120	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_S4	0.221	78	0.271	+4.9	95	0.287	+6.6	101	fmt	fmt	fmt	-	-	-	-	-	-
C4_L1r_S1_S3_S4	0.203	71	0.251	+4.8	88	0.281	+7.8	99	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1	0.136	48	0.153	+1.7	54	0.173	+3.7	61	fmt	fmt	fmt	-	-	-	-	-	-
C_X1_S1	0.153	54	0.200	+4.7	70	0.244	+9.1	86	fmt	fmt	fmt	-	-	-	-	-	-
C_S2_X1	0.209	73	0.246	+3.7	86	0.272	+6.3	95	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1m	0.158	55	0.173	+1.5	61	0.211	+5.3	74	fmt	fmt	fmt	-	-	-	-	-	-
C_S1_X1s	0.206	72	0.259	+5.3	91	0.293	+8.7	103	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_X1	0.167	59	0.204	+3.7	72	0.217	+5.0	76	fmt	fmt	fmt	-	-	-	-	-	-

Table 28. **Merging, Llama-3.1-8B.** Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	acc	%	merge_ties	acc	Δ	merge_dare_ties	acc	Δ	merge_dare_linear	acc	Δ	l0merge_ties	acc	Δ	l0merge_dare_ties	acc	Δ
L0	0.259	100	0.398	+14.0	154	0.462	+20.3	179	0.292	+3.4	113	-	-	-	-	-	-
L1b	0.222	86	0.336	+11.4	130	0.386	+16.4	150	0.253	+3.1	98	-	-	-	-	-	-
L1r	0.242	94	0.348	+10.6	135	0.398	+15.6	154	0.234	-0.8	91	-	-	-	-	-	-
L2	0.215	83	0.353	+13.8	137	0.396	+18.1	153	0.246	+3.1	95	-	-	-	-	-	-
S1	0.213	83	0.335	+12.2	130	0.412	+19.9	160	0.256	+4.3	99	-	-	-	-	-	-
S2	0.208	80	0.361	+15.3	140	0.437	+23.0	169	0.256	+4.9	99	-	-	-	-	-	-
X1	0.132	51	0.217	+8.5	84	0.266	+13.4	103	fmt	fmt	fmt	-	-	-	-	-	-
C_L1b_S1	0.152	59	0.252	+10.1	98	0.307	+15.6	119	0.211	+6.0	82	-	-	-	-	-	-
C_L1r_S1	0.162	63	0.273	+11.1	106	0.328	+16.6	127	0.206	+4.4	80	-	-	-	-	-	-
C_S1_L1r	0.159	62	0.247	+8.9	96	0.306	+14.8	119	0.200	+4.2	78	-	-	-	-	-	-
C_L2_S4	0.206	80	0.311	+10.5	120	0.383	+17.8	149	0.226	+2.0	88	-	-	-	-	-	-
C_L1r_S3	0.198	77	0.334	+13.5	129	0.387	+18.9	150	0.222	+2.3	86	-	-	-	-	-	-
C_S4_S3	0.214	83	0.357	+14.3	138	0.437	+22.3	169	0.255	+4.1	99	-	-	-	-	-	-
C3_L1r_S3_S4	0.184	71	0.306	+12.2	119	0.370	+18.6	143	0.199	+1.6	77	-	-	-	-	-	-
C3_S1_S3_S4	0.157	61	0.308	+15.1	119	0.377	+22.0	146	0.220	+6.2	85	-	-	-	-	-	-
C3_L1r_S1_S4	0.166	64	0.260	+9.4	101	0.319	+15.3	124	0.183	+1.7	71	-	-	-	-	-	-
C4_L1r_S1_S3_S4	0.134	52	0.235	+10.1	91	0.310	+17.6	120	0.179	+4.5	69	-	-	-	-	-	-
C_L1r_X1	0.100	39	0.147	+4.6	57	0.184	+8.3	71	fmt	fmt	fmt	-	-	-	-	-	-
C_X1_S1	0.130	50	0.233	+10.3	90	0.296	+16.6	115	fmt	fmt	fmt	-	-	-	-	-	-
C_S2_X1	0.169	66	0.263	+9.4	102	0.317	+14.8	123	fmt	fmt	fmt	-	-	-	-	-	-
C_L1r_X1m	0.114	44	0.162	+4.8	63	0.217	+10.3	84	0.126	+1.2	49	-	-	-	-	-	-
C_S1_X1s	0.198	77	0.287	+8.9	111	0.349	+15.1	135	fmt	fmt	fmt	-	-	-	-	-	-
C3_L1r_S1_X1	0.105	41	0.170	+6.6	66	0.215	+11.0	83	fmt	fmt	fmt	-	-	-	-	-	-

Table 29. **Merging, StarCoder2-15B**. Weight-space merges of the per-transform specialists (TIES, DARE-TIES, DARE-linear) and two merges anchored on the clean-code adapter, on every condition and stack that has them. Same three numbers as the master grid.

Condition	untuned base		TIES			DARE-TIES			DARE-linear			L0-anch. TIES			L0-anch. DARE-TIES		
	acc	%	merge_ties	Δ	%	merge_dare_ties	Δ	%	merge_dare_linear	Δ	%	l0merge_ties	Δ	%	l0merge_dare_ties	Δ	%
L0	fnt	fnt	0.484	fnt	fnt	0.539	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
L1b	fnt	fnt	0.366	fnt	fnt	0.426	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
L1r	fnt	fnt	0.414	fnt	fnt	0.476	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
L2	fnt	fnt	0.401	fnt	fnt	0.466	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
S1	fnt	fnt	0.451	fnt	fnt	0.487	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
S2	fnt	fnt	0.440	fnt	fnt	0.518	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
X1	fnt	fnt	0.283	fnt	fnt	0.343	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C_L1b_S1	fnt	fnt	0.321	fnt	fnt	0.354	fnt	fnt	0.063	fnt	fnt	fnt	-	-	-	-	-
C_L1r_S1	fnt	fnt	0.344	fnt	fnt	0.389	fnt	fnt	0.047	fnt	fnt	fnt	-	-	-	-	-
C_S1_L1r	fnt	fnt	0.346	fnt	fnt	0.387	fnt	fnt	0.041	fnt	fnt	fnt	-	-	-	-	-
C_L2_S4	fnt	fnt	0.381	fnt	fnt	0.438	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C_L1r_S3	fnt	fnt	0.390	fnt	fnt	0.459	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C_S4_S3	fnt	fnt	0.442	fnt	fnt	0.506	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C3_L1r_S3_S4	fnt	fnt	0.377	fnt	fnt	0.436	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C3_S1_S3_S4	fnt	fnt	0.393	fnt	fnt	0.464	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C3_L1r_S1_S4	fnt	fnt	0.332	fnt	fnt	0.376	fnt	fnt	0.045	fnt	fnt	fnt	-	-	-	-	-
C4_L1r_S1_S3_S4	fnt	fnt	0.301	fnt	fnt	0.358	fnt	fnt	0.026	fnt	fnt	fnt	-	-	-	-	-
C_L1r_X1	fnt	fnt	0.215	fnt	fnt	0.259	fnt	fnt	0.058	fnt	fnt	fnt	-	-	-	-	-
C_X1_S1	fnt	fnt	0.295	fnt	fnt	0.371	fnt	fnt	0.067	fnt	fnt	fnt	-	-	-	-	-
C_S2_X1	fnt	fnt	0.341	fnt	fnt	0.403	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C_L1r_X1m	fnt	fnt	0.252	fnt	fnt	0.317	fnt	fnt	0.065	fnt	fnt	fnt	-	-	-	-	-
C_S1_X1s	fnt	fnt	0.378	fnt	fnt	0.427	fnt	fnt	fnt	fnt	fnt	fnt	-	-	-	-	-
C3_L1r_S1_X1	fnt	fnt	0.241	fnt	fnt	0.286	fnt	fnt	0.058	fnt	fnt	fnt	-	-	-	-	-

Table 30. **RQ1: routing and merging on stacks of seen transforms**. Paired differences in points on the ten stacks (six depth-2, four depth-3/4), CodeLlama-7B, 394 programs common to every cell; the last column repeats the contrasts on the four stacks that contain the unseen family (319 programs; merges other than DARE-TIES have no cells there). Routing is worth about a point over a random gate and nothing over breadth; the mixture itself carries the gain (mole_uniform – base); a hard argmax router is indistinguishable from the soft one. The best merge (DARE-TIES) matches the clean-code control and the other four fall below it. Bold intervals exclude zero. Every arm ran through the mixture engine (obtune.mole.eval_mole); an earlier run through vLLM had written the untuned model under the routing arms’ names and was withdrawn.

Contrast	all ten seen	depth 2	depth 3–4	unseen inside
<i>Routing</i>				
mole_router – mole_random	+1.34 [+0.48, +2.17]	+1.11 [+0.28, +1.93]	+1.67 [+0.47, +2.86]	–0.03 [–0.99, +0.84]
mole_router – mole_uniform	+1.40 [+0.52, +2.25]	+1.17 [+0.31, +2.02]	+1.74 [+0.53, +2.90]	+0.10 [–0.84, +0.99]
mole_hardrouter – mole_router	+0.03 [–0.13, +0.19]	–0.03 [–0.24, +0.18]	+0.13 [–0.11, +0.38]	–0.13 [–0.44, +0.18]
mole_router – mono_all	–0.28 [–1.57, +1.09]	–0.06 [–1.38, +1.30]	–0.61 [–2.10, +0.91]	+2.88 [+1.33, +4.41]
mole_uniform – base	+19.79 [+17.40, +22.23]	+19.16 [+16.77, +21.58]	+20.72 [+18.16, +23.31]	+14.98 [+12.81, +17.29]
<i>Merging</i>				
merge_dare_ties – tuned_L0	+0.52 [–0.15, +1.20]	+0.61 [–0.16, +1.41]	+0.38 [–0.47, +1.25]	+0.86 [–0.05, +1.78]
merge_ties – tuned_L0	–5.50 [–6.86, –4.13]	–5.19 [–6.59, –3.77]	–5.95 [–7.47, –4.44]	–
merge_dare_linear – tuned_L0	–20.56 [–22.88, –18.25]	–20.60 [–23.00, –18.22]	–20.49 [–22.86, –18.15]	–
l0merge_ties – tuned_L0	–3.14 [–4.21, –2.12]	–2.81 [–3.98, –1.67]	–3.64 [–4.91, –2.41]	–
l0merge_dare_ties – tuned_L0	–1.74 [–2.57, –0.91]	–1.40 [–2.30, –0.47]	–2.24 [–3.26, –1.23]	–
<i>Breadth, for reference</i>				
mono_all – tuned_L0	+3.67 [+2.02, +5.24]	+3.34 [+1.61, +5.04]	+4.15 [+2.35, +5.91]	–1.49 [–3.08, +0.08]

Table 31. **Weight-space merging on the panel.** The best merge (DARE-TIES of the clean adapter and the five specialists) against the clean-code control, breadth and the anchored objective, pooled over the six seen stacks and the four stacks containing the unseen family; breadth and anchoring against the control in the last two columns. Points, paired on the programs common to every cell, program-clustered bootstrap. Bold intervals exclude zero. Pre-registered per model: merging does not beat the control on seen stacks (CONFIRMED iff $ci_{hi} \leq +1$) and stays below breadth ($ci_{hi} < 0$).

Stacks	merge-clean	merge-mono	merge-cons	mono-clean	cons-clean
CodeLlama-7B					
seen (6)	+0.83 [+0.01, +1.63]	-2.80 [-4.43, -1.19]	-4.01 [-5.29, -2.70]	+3.63 [+1.96, +5.27]	+4.84 [+3.49, +6.21]
unseen inside (4)	+0.86 [-0.05, +1.78]	+2.35 [+0.63, +3.97]	-1.39 [-2.72, -0.05]	-1.49 [-3.08, +0.08]	+2.25 [+0.86, +3.61]
CodeLlama-13B					
seen (6)	+1.67 [+0.83, +2.51]	-1.40 [-2.98, +0.12]	-2.41 [-3.85, -1.05]	+3.07 [+1.55, +4.63]	+4.08 [+2.72, +5.40]
unseen inside (4)	+1.75 [+0.78, +2.80]	+2.48 [+0.89, +4.08]	+0.08 [-1.23, +1.39]	-0.73 [-2.23, +0.76]	+1.67 [+0.45, +2.82]
Llama-3.1-8B					
seen (6)	+2.31 [+1.27, +3.38]	-0.85 [-2.41, +0.72]	-2.80 [-4.26, -1.35]	+3.16 [+1.52, +4.84]	+5.11 [+3.63, +6.58]
unseen inside (4)	+2.56 [+1.49, +3.61]	+3.09 [+1.54, +4.60]	+0.81 [-0.47, +2.12]	-0.52 [-2.07, +1.02]	+1.75 [+0.39, +3.11]
StarCoder2-15B					
seen (6)	-0.69 [-1.57, +0.13]	-3.27 [-4.67, -1.86]	-5.73 [-7.05, -4.47]	+2.57 [+1.28, +3.86]	+5.04 [+3.82, +6.26]
unseen inside (4)	+1.15 [+0.08, +2.17]	+1.70 [+0.05, +3.32]	+0.39 [-0.92, +1.80]	-0.55 [-2.12, +1.02]	+0.76 [-0.47, +2.02]
Gemma-3-12B					
seen (6)	+1.27 [+0.32, +2.22]	-2.61 [-4.13, -1.11]	-3.91 [-5.28, -2.52]	+3.88 [+2.18, +5.64]	+5.17 [+3.69, +6.64]
unseen inside (4)	+0.42 [-0.55, +1.36]	+1.83 [+0.39, +3.40]	-0.18 [-1.57, +1.20]	-1.41 [-3.01, +0.13]	+0.60 [-0.78, +1.96]
CodeGemma-7B					
seen (6)	+0.44 [-0.48, +1.36]	-3.83 [-5.34, -2.32]	-3.59 [-5.08, -2.24]	+4.27 [+2.83, +5.63]	+4.03 [+2.67, +5.46]
unseen inside (4)	-0.10 [-1.02, +0.81]	-0.13 [-1.67, +1.44]	+0.92 [-0.47, +2.35]	+0.03 [-1.55, +1.57]	-1.02 [-2.38, +0.29]
Granite-3.1-8B					
seen (6)	+1.57 [+0.76, +2.46]	-3.23 [-4.92, -1.52]	-2.32 [-4.01, -0.61]	+4.80 [+3.05, +6.53]	+3.89 [+2.12, +5.67]
unseen inside (4)	+2.09 [+1.10, +3.09]	+2.20 [+0.50, +3.84]	+2.98 [+1.28, +4.65]	-0.10 [-1.78, +1.62]	-0.89 [-2.59, +0.89]

Table 32. **The direction ratio: how much of a forward gain survives the task being run backwards.** Forward is output prediction; backward is input prediction on the same programs, graded by execution. No adapter is trained on the backward task, so any backward gain is transferred program understanding rather than task fitting. $DR < 0$ means the arm bought forward accuracy by *losing* backward competence. Every adaptation method that fits the forward task directly is negative; only anchoring and family exposure are not. Both directions use the same 412 programs and the same six conditions. Bold intervals exclude zero; the ratio is descriptive and is read with its components.

Method	Arm	Forward gain	Backward gain	DR
Format-only control	formatonly	+1.47 [+0.90, +2.10]	-0.44 [-0.81, -0.08]	-0.30
Clean-code tuning	tuned_L0	+19.34 [+16.87, +21.81]	-1.60 [-3.41, +0.35]	-0.08
Training breadth	mono_all	+19.46 [+16.68, +22.01]	-0.89 [-2.86, +1.04]	-0.05
Paired consistency (ours)	cons_lam3	+20.88 [+18.07, +23.49]	+0.62 [-1.56, +2.81]	+0.03
Family exposure	tuned_X1	+17.35 [+14.98, +19.73]	+1.63 [+0.11, +3.20]	+0.09

Table 33. **RQ4: the divergence result across the panel**, on stacks containing the unseen family. cons-mono is anchoring’s advantage over breadth; fam is an arm trained on the unseen family’s trainable sibling, included because family exposure is the largest effect in the table and replicates on every model. Bold intervals exclude zero.

Model	Lineage	cons-mono	cons-clean	fam-clean
CodeLlama-7B	Meta	+3.84 [+1.90, +5.70]	+2.13 [+0.66, +3.60]	+5.93 [+4.10, +7.78]
CodeLlama-13B	Meta	+3.22 [+1.74, +4.65]	+2.02 [+0.58, +3.53]	+5.97 [+4.11, +7.91]
CodeLlama-34B	Meta	+3.95 [+2.13, +5.81]	+3.10 [+1.63, +4.58]	+4.19 [+2.48, +5.94]
Llama-3.1-8B	Meta	+2.87 [+1.20, +4.62]	+1.86 [+0.27, +3.53]	+6.67 [+4.58, +8.71]
StarCoder2-15B	BigCode	+1.36 [−0.08, +2.71]	+0.35 [−1.13, +1.86]	+3.64 [+1.90, +5.35]
Gemma-3-12B	Google	+2.44 [+0.97, +3.99]	−0.08 [−1.67, +1.51]	+5.35 [+3.21, +7.37]
CodeGemma-7B	Google	−1.59 [−3.26, +0.12]	−2.40 [−4.11, −0.85]	+5.23 [+3.25, +7.21]
Granite-3.1-8B	IBM	−1.47 [−2.99, +0.00]	−1.86 [−3.80, +0.12]	+7.29 [+5.19, +9.65]

Table 34. **Every intervention against every kind of obfuscation, across the eight-model panel**. For each system and direction, Δ is the mean change in points against the untuned model and % the mean accuracy as a percentage of the untuned model’s clean-code accuracy, over the cells that exist and pass the format gate (in parentheses: the number of cells; after the semicolon, the number gated as *fnt*). Systems: ICL = base_1shot, clean LoRA = tuned_L0, breadth = mono_all, anchored = cons_lam3, family = tuned_X1. *seen* = L1b, L1r, L2, S1, S2; *seen stacks* = the six depth-2 and four depth-3/4 stacks of seen transforms; *unseen stacks* = the three depth-2 and one depth-3 stacks with the X1 family inside. Backward cells survive the gate almost only on CodeLlama-7B, whose grid (Table 17) is the one fully interpretable backward read; the per-model grids for the whole panel are in Appendix B.

System	measure	clean	seen	unseen family	seen stacks	unseen stacks
ICL	fwd Δ pts	+3.8 (3; 5 fnt)	+3.2 (16; 24 fnt)	−	−	−
	fwd % of clean	115% (3; 5 fnt)	97% (16; 24 fnt)	−	−	−
clean LoRA	fwd Δ pts	+19.0 (5; 3 fnt)	+18.5 (30; 10 fnt)	+13.7 (7; 1 fnt)	+16.7 (60; 20 fnt)	+11.5 (28; 4 fnt)
	fwd % of clean	175% (5; 3 fnt)	152% (25; 15 fnt)	102% (5; 3 fnt)	129% (50; 30 fnt)	95% (20; 12 fnt)
	bwd Δ pts	−0.4 (7; 1 fnt)	−0.3 (28; 12 fnt)	+0.2 (5; 3 fnt)	+0.8 (54; 26 fnt)	−0.8 (20; 12 fnt)
	bwd % of clean	101% (7; 1 fnt)	97% (32; 8 fnt)	79% (5; 3 fnt)	90% (63; 17 fnt)	79% (21; 11 fnt)
breadth	fwd Δ pts	+17.0 (5; 3 fnt)	+19.7 (30; 10 fnt)	+10.7 (7; 1 fnt)	+20.9 (60; 20 fnt)	+10.9 (28; 4 fnt)
	fwd % of clean	167% (5; 3 fnt)	157% (25; 15 fnt)	91% (5; 3 fnt)	146% (50; 30 fnt)	93% (20; 12 fnt)
	bwd Δ pts	−2.2 (6; 2 fnt)	−1.6 (26; 14 fnt)	−1.1 (4; 4 fnt)	−0.0 (50; 30 fnt)	−4.3 (15; 17 fnt)
	bwd % of clean	95% (6; 2 fnt)	91% (29; 11 fnt)	73% (4; 4 fnt)	88% (52; 28 fnt)	64% (17; 15 fnt)
anchored	fwd Δ pts	+17.6 (5; 3 fnt)	+21.0 (30; 10 fnt)	+13.1 (7; 1 fnt)	+22.0 (60; 20 fnt)	+12.8 (28; 4 fnt)
	fwd % of clean	169% (5; 3 fnt)	161% (25; 15 fnt)	99% (5; 3 fnt)	148% (50; 30 fnt)	98% (20; 12 fnt)
	bwd Δ pts	−4.4 (6; 2 fnt)	−5.1 (22; 18 fnt)	−2.6 (4; 4 fnt)	−4.2 (39; 41 fnt)	−3.4 (9; 23 fnt)
	bwd % of clean	89% (6; 2 fnt)	86% (26; 14 fnt)	75% (5; 3 fnt)	81% (49; 31 fnt)	75% (9; 23 fnt)
family	fwd Δ pts	+18.5 (5; 3 fnt)	+17.8 (30; 10 fnt)	+20.1 (7; 1 fnt)	+17.5 (60; 20 fnt)	+16.8 (28; 4 fnt)
	fwd % of clean	173% (5; 3 fnt)	150% (25; 15 fnt)	128% (5; 3 fnt)	132% (50; 30 fnt)	116% (20; 12 fnt)
	bwd Δ pts	−1.4 (7; 1 fnt)	−2.8 (31; 9 fnt)	+1.6 (5; 3 fnt)	−1.0 (58; 22 fnt)	−0.9 (21; 11 fnt)
	bwd % of clean	96% (7; 1 fnt)	87% (33; 7 fnt)	80% (6; 2 fnt)	84% (59; 21 fnt)	76% (23; 9 fnt)